

## Research paper

# Multi-objective decision-making for flexible design and planning of oxygen production facilities in an uncertain lunar environment

Kosuke Ikeya <sup>a,b</sup>, Michel-Alexandre Cardin <sup>b,\*</sup>, Jan Cilliers <sup>a</sup>, Stanley O. Starr <sup>a</sup>, Kathryn Hadler <sup>a,c</sup>

<sup>a</sup> Department of Earth Science and Engineering, Imperial College London, Royal School of Mines, Prince Consort Road, London, SW7 2AZ, United Kingdom

<sup>b</sup> Dyson School of Design Engineering, Imperial College London, 25 Exhibition Road, London, SW7 2AZ, United Kingdom

<sup>c</sup> ESRC (European Space Resources Innovation Centre), LIST (Luxembourg Institute of Science and Technology), 28 Avenue des Hauts-Fourneaux, Esch-sur-Alzette, L-4362, Luxembourg

## ARTICLE INFO

## Keywords:

In-situ resource utilization  
Oxygen extraction  
Carbothermal reduction  
Icy regolith  
Bayesian inference  
Multi-objective decision making  
Decision analysis  
Real options analysis

## ABSTRACT

In-Situ Resource Utilization (ISRU), the act of utilizing local extraterrestrial resources, such as the lunar regolith, has been proposed and researched extensively to establish a self-sustained human presence in space. Oxygen production from lunar resources has gained significant attention, especially with growing interest in lunar exploration. Previous studies have recognized the uncertainties associated with the lunar environment and operations of ISRU plants, such as resource content, and their effects on ISRU plant performance. These studies, however, often overlook the fact that knowledge about uncertainty can be continuously updated through information-gathering. This paper analyzes the design and planning of ISRU facilities as a multi-objective sequential decision-making problem under uncertainty. Inspired by the value of information analyses commonly used in the terrestrial resource extraction industry, this research employs multi-objective decision tree analysis to examine different lunar ISRU plant designs and deployment strategies, identifying potential risk management trade-offs in oxygen production. Monte Carlo simulations reveal the effects of various uncertain parameters on the performance of each ISRU plant, while Bayesian inference updates these uncertain parameters. A case study of the deployment of pilot and full-scale ISRU plants in the lunar southern polar region reveals trade-offs between design and planning decisions. Four different oxygen production architectures are considered in this case study: carbothermal reduction of dry regolith, water extraction from icy regolith, and hybrid of these technologies connecting them in parallel and series. The results suggest well-balanced performances of deploying a parallel or series hybrid pilot plant regarding all three metrics: oxygen yield, mass payback, and mass-energy efficiency. On the other hand, deploying water extraction architecture for both pilot and full-scale plants may work best for mass payback and mass-energy efficiency, while it also shows significantly smaller oxygen yield. The study also highlights the flexibility added by deploying a hybrid lunar ISRU pilot plant, which allows decision-makers to adjust their strategies as needed.

## 1. Introduction

More than 50 years after the Apollo 17 astronauts left the Moon, humans are finally ready to return there. The Artemis Program [1] led by NASA and its partner institutions, such as ESA and JAXA, tries to establish a community on and around the Moon. Concurrently, China and Russia also aim to build a Moon base, the International Lunar Research Station [2], to promote international scientific cooperation on the Moon. These goals of enabling longer-term or even permanent human presence in cislunar space contrast with a few days' stays by Apollo astronauts. To support sustained human presence on the Moon

and beyond, the concept of using resources in space, In-Situ Resource Utilization (ISRU), also known as space resource utilization, has been proposed and is being increasingly researched.

Among resources on the Moon, lunar regolith and water ice in the polar regions have gathered significant attention for their potential to support oxygen extraction and production. Oxygen is one of the most desirable products through lunar ISRU due to its extensive applicability for astronauts' life support and spacecraft propellant [3].

Various methods have been proposed to extract oxygen from lunar regolith [3,4] and demonstrated on a laboratory scale [5–7] such as

\* Corresponding author.

E-mail address: [m.cardin@imperial.ac.uk](mailto:m.cardin@imperial.ac.uk) (M.-A. Cardin).

<https://doi.org/10.1016/j.actaastro.2025.07.014>

Received 31 March 2025; Received in revised form 29 May 2025; Accepted 8 July 2025

Available online 24 July 2025

0094-5765/© 2025 The Authors. Published by Elsevier Ltd on behalf of IAA. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

**Nomenclature**

|                          |                                                                 |
|--------------------------|-----------------------------------------------------------------|
| <b>A</b>                 | set of actions                                                  |
| <b>A</b>                 | set of action branches                                          |
| <b>a</b>                 | action branch                                                   |
| <b>C</b>                 | set of chance nodes                                             |
| <b>C</b>                 | chance node                                                     |
| <b>D</b>                 | set of decision nodes                                           |
| <b>D</b>                 | decision node                                                   |
| <b>E</b>                 | set of states of nature                                         |
| <b>E</b>                 | set of chance branches                                          |
| <b>e</b>                 | chance branch                                                   |
| <b>E<sub>total</sub></b> | total energy                                                    |
| <b>f(·)</b>              | function representing performance measures of a scenario        |
| <b>g(·)</b>              | function representing a system                                  |
| <b>k</b>                 | number of chance branches from a chance node                    |
| <b>L</b>                 | set of integers corresponding to objectives                     |
| <b>l</b>                 | number of objectives to be optimized                            |
| <b>m<sub>O2</sub></b>    | total produced oxygen mass                                      |
| <b>m<sub>reg</sub></b>   | total excavated regolith mass                                   |
| <b>m<sub>sys</sub></b>   | total landed mass                                               |
| <b>n</b>                 | number of action branches from a decision node                  |
| <b>P</b>                 | Pareto optimal set of decision strategies                       |
| <b>p</b>                 | general probability                                             |
| <b>p(·   ·)</b>          | conditional probability                                         |
| <b>r</b>                 | performance measure vector                                      |
| <b>s</b>                 | system state                                                    |
| <b>T</b>                 | set of terminal nodes                                           |
| <b>T</b>                 | decision tree                                                   |
| <b>T</b>                 | terminal node                                                   |
| <b>x</b>                 | deterministic input parameters for a system                     |
| <b>y</b>                 | output of a system                                              |
| <b>θ</b>                 | uncertain input parameters for a system                         |
| <b>Θ</b>                 | set of uncertain parameters generated by the Monte Carlo method |

**Abbreviations**

|        |                              |
|--------|------------------------------|
| ConOps | concept of operations        |
| CR     | carbothermal reduction       |
| ISRU   | in-situ resource utilization |
| PDF    | probability density function |
| PEL    | peak of eternal light        |
| PH     | parallel hybrid              |
| PSR    | permanently shadowed region  |
| LOX    | liquid oxygen                |
| SH     | series hybrid                |
| SOXE   | solid oxide electrolysis     |
| WE     | water extraction             |

hydrogen reduction of ilmenite [7–9] and carbothermal reduction [5,6,10–12] followed by water electrolysis. Oxygen can also be extracted by electrolyzing water from water ice available in polar regions [13–15]. Identifying the most promising plant architecture among these methods is crucial for successful and economically viable lunar ISRU [3,4,16]. To identify such an architecture, previous studies have compared the mass [17–20], power [18,20], and monetary costs [17] of different plant architectures.

While these studies identified advantages and disadvantages of each ISRU architecture, their assessments depend on deterministic assumptions in key parameters, such as resource content in excavated regolith. The potential impact of uncertainty on lunar ISRU Concept of Operations (ConOps) and system design has been assessed through Monte Carlo simulations considering uncertainty in lunar resource content and ISRU operations [20,21].

Despite considerable progress, these past studies did not consider the potential benefits of continuously updating our knowledge of the lunar environment as ISRU facilities are deployed and operated. For instance, demonstration missions on the Moon and pilot ISRU plant operations can help improve our understanding of resource availability, which may impact decisions in later stages, e.g., full-scale plant deployment. Overlooking the early-stage deployment of ISRU and its relation to later-stage operations is a critical gap that needs to be addressed for more effective and efficient lunar ISRU system designs and operations.

To address this research gap, this paper applies multi-objective Bayesian decision tree analyses [22] to determine the benefits and risks of different lunar ISRU designs and planning. Considering multiple objectives is essential in lunar ISRU deployment to reveal possible risks that can be inadvertently dismissed in single-objective decision analyses. For example, decisions solely based on minimizing costs may miss other possible solutions that fare better in terms of energy consumption or resource extraction rate. To examine the effects of uncertainty on the performance of each plant architecture, Monte Carlo simulations are conducted. Uncertain parameter distributions for these simulations are updated through Bayesian inference. The results from the Monte Carlo simulations are, then, integrated into multi-objective the decision tree.

Our proposed approach is demonstrated through a case study of lunar in-situ Liquid O<sub>2</sub> (LOX) production plants in the lunar southern polar region. This case study deploys first a pilot ISRU plant, which enables information gathering on ISRU operations and lunar resources, followed by a full-scale plant. Although this case study focuses on the design and planning of lunar ISRU systems, this method can be applied to other ISRU locations such as Mars and near-Earth objects.

The rest of the paper is organized as follows. Related works are briefly explored in Section 2 clarifying the research gap. Section 3 describes the methodology, multi-objective decision analysis, Monte Carlo simulations, and Bayesian inference, employed to quantify the performance of different design and planning strategies. Section 4 demonstrates the usefulness of the proposed method through a case study of early-stage lunar ISRU plant deployment in the southern polar region. Key findings and limitations are summarized in Section 5, also highlighting potential directions for future research. Finally, Section 6 concludes the paper.

## 2. Related works and research gap

### 2.1. Lunar oxygen production processes

Various methods to extract oxygen on the Moon have been proposed for life support and to produce propellant beyond Earth. Oxygen can be extracted from lunar dry regolith through, e.g., hydrogen reduction of ilmenite [7–9], carbothermal reduction [5,6,10–12], molten regolith electrolysis [23,24], and molten salt electrolysis [25]. For icy regolith, the water extraction method can be categorized into three main approaches: heating excavated regolith [13,14], using heated drilling rods [26–28], and building a heat dome [15,29]. Schlüter and Cowley [3] provide a comprehensive review of such technologies to extract oxygen on the Moon.

These technologies have been discussed and compared from various perspectives [4,17–20,30]. For instance, Taylor and Carrier [4] evaluated 20 different technologies (e.g., ilmenite reduction with hydrogen, molten silicate electrolysis, and vapor pyrolysis) from multiple metrics such as technology readiness level, complexity of the process, and feedstock availability. They evaluated each technology scoring 1–10 for

each criterion and identified more feasible ones by comparing the total scores. Although their evaluation is comprehensive, it needs updating to reflect the latest progress in ISRU technologies and newly proposed processes such as water extraction from icy regolith.

Chen et al. [17] made an ISRU mass and power estimation model to optimize the design of ISRU for crewed Mars missions. This model includes all essential subsystems from excavator to storage. To integrate this model into a crewed campaign to Mars, the model is rather high-level and low-fidelity. Conversely, Guerrero-Gonzalez and Zabel [18] developed more detailed mass and power estimation models for different lunar ISRU architectures for both oxygen and metal production. Through their analysis, the significant sensitivity of the hydrogen reduction process followed by carbonylation to the mineral composition of regolith was revealed.

Although these models are useful for comparing the system mass and required power of lunar oxygen production architectures, they rely on deterministic assumptions regarding critical parameters such as resource availability. As highlighted by Cilliers et al. [31], consideration of uncertainty inherent in lunar ISRU is crucial. Cilliers et al. [21] conducted Monte Carlo simulations to examine the effect of uncertainty on lunar resources and ISRU operations. Their study indicates that ignoring uncertainty can lead to the risk of not meeting the target production or changes to ConOps that may be very costly.

Ikeya et al. [20] further extended this analysis for other lunar ISRU architectures: carbothermal reduction, water extraction from icy regolith, and hybrid architecture integrating these two technologies. The result of this study clarified the effect of uncertainty in lunar polar ice availability on the system mass, required power, and ConOps of water extraction and the hybrid architectures. However, while recognizing the importance of uncertainty in lunar ISRU, this study did not consider the impact of updating this uncertainty over time through information gathering and its effect on the decision-making process.

## 2.2. Multistage operations of engineering systems

The act of planning, deploying, and operating ISRU plants on the Moon is likely to be analogous to that of terrestrial resource (e.g., minerals and oil) extraction [32,33]. The development and deployment of these engineering systems have been studied in the process industry [34–36]. When novel and developing technologies are involved, a multistage approach, from technology testing to full-scale production, helps reduce uncertainty related to resource availability and technologies, both contributing to higher likelihood of success [34,37]. Running pilot-scale operations enables better-informed decisions for full-scale plants, for example by enabling selection of appropriate resource extraction technology, given available resources, and technological efficiency. Testing may also include laboratory-scale testing, on-site demonstration, and pilot plant operations before full-scale deployment [34].

Despite the recognized importance of pilot-scale ISRU [38], the design and operations of lunar ISRU pilot plants have not been studied thoroughly. Linne et al. [14] and Dottori et al. [39] estimated the system mass and required power for pilot plants for water extraction from icy regolith and carbothermal reduction of dry regolith, respectively. However, these studies solely focused on the design of pilot plants. The link between pilot plant performance and decision-making for full-scale plants remains unexplored.

Some space logistics studies have considered the staged deployment of ISRU systems [17,40,41]. Their approaches could be applied to the study of pilot and full-scaled ISRU plants. For instance, Takubo et al. [41] recognized uncertainty in production rate and optimized the ISRU system deployment through two to five launches to the lunar surface. Extending these studies to include uncertainty reduction by pilot plant operations has not yet been done.

In terrestrial applications, the staged deployment of engineering systems has often been solved as sequential decision problems [37,

42–44]. Sequential decision-making involves a series of decisions and information-gathering steps. Each decision is based on the current information, which is influenced by previous decisions.

Various methods to solve a sequential decision problem [45] include Markov decision processes [46,47], multistage stochastic programming [48], model predictive control [49,50], and decision tree analyses [37,42,44,51]. Decision tree analysis, proposed by Morgan and Sonquist [51], is one of the simplest approaches to solving a sequential decision problem. A decision tree consists of a sequence of stages made of “decision” and “chance” nodes, typically spread out over time and/or space. The value of each decision is calculated as an expected value over all possible outcomes at any given stage, using pre-assigned probabilities. A backward induction – or rollback – process from dynamic programming theory helps identify the best decision in each stage, until time zero. Due to their simplicity and clarity, decision trees are widely used for solving sequential decision problems and to support better communication between relevant stakeholders and decision-makers.

## 2.3. Value of information and value of flexibility

In both terrestrial and space resource extraction, the trade space between exploration and exploitation is critical [52]. In terrestrial applications, Value of Information (VoI) measures the (typically financial) value gained by additional testing to gather more information. This metric has often been used to compare relevant design and planning decisions for resource extraction infrastructures [37,44,53]. For instance, in mining, where ore grade varies spatially, constantly updated information on a mining site can be economically beneficial [53]. Chen et al. [37] conducted a decision tree analysis to calculate the VoI obtained through a pilot project for reservoir management. Samis and Steen [44] used VoI to evaluate pilot mining projects with new technologies and suggested pilot projects are valuable for risk management.

Similarly, the Value of Flexibility (VoF) has been calculated in many engineering applications [42,54]. While VoI focuses on reducing uncertainty by gaining more information, VoF measures the value gained by exercising strategic and tactical flexibility carefully designed in engineering systems, so systems can adapt, change, and reconfigure easily in operations as uncertainty unfolds [55,56].

Both VoI and VoF are often discussed alongside Real Options Analysis (ROA) [57,58], which provides the “right, but not the obligation” [59] to change a project responding to uncertainty. For example, Davis and Samis [58] used ROA to assess the value of exploration activities in the mining industry, considering geological and metal price uncertainty. The concepts of VoI and VoF naturally emerge from ROA theory, though in this study the focus is on a system’s ability to adapt, and what is needed to enable this flexibility from an engineering and design standpoint — as opposed to just economic quantification as in standard ROA.

## 2.4. Contribution

The key contribution of this paper is an assessment of different lunar ISRU designs and planning considering the impact of information gathering. Future operations of lunar ISRU will likely involve multiple stages, from demonstration to full-scale plant production. Information gathered in early stages may be used to inform better decisions in subsequent stages. Given this dynamic nature, this paper applies a sequential decision-making approach to address the problem of how to best deploy lunar ISRU capacity in the future.

The method employed in this paper is multi-objective decision tree analysis, chosen for its simplicity. Unlike a single-objective approach, the multi-objective approach enables visualizing trade-offs across multiple metrics. Monte Carlo simulations assign probabilities to the decision tree, and Bayesian inference updates probability distributions to

**Table 1**

Comparison of methods in past studies and this paper. Key nine papers in decision-making on systems design, planning, and/or operation are summarized here.

| Reference                                             | Method                                                                      | Application                                              | Decision-making |            | Uncertainty   |        | VoI/VoF analysis |
|-------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------|-----------------|------------|---------------|--------|------------------|
|                                                       |                                                                             |                                                          | Muti-objective  | Sequential | Consideration | Update |                  |
| <b><i>Space logistics papers</i></b>                  |                                                                             |                                                          |                 |            |               |        |                  |
| Chen et al. [17]                                      | ISRU modeling + space logistics network optimization                        | Crewed Mars exploration campaign                         |                 | ✓          |               |        |                  |
| Takubo et al. [41]                                    | Hierarchical reinforcement learning + space logistics network optimization  | Crewed lunar exploration campaign                        |                 | ✓          | ✓             |        |                  |
| <b><i>Real Options and flexibility papers</i></b>     |                                                                             |                                                          |                 |            |               |        |                  |
| Zuluaga and Sánchez-Silva [43]                        | Stochastic programming + Real Options                                       | Infrastructure maintenance                               |                 | ✓          | ✓             |        | ✓                |
| Samis and Steen [44]                                  | Decision tree + Bayesian inference                                          | Terrestrial mining pilot project                         |                 |            | ✓             | ✓      | ✓                |
| Cardin et al. [48]                                    | Decision rule + multistage stochastic programming                           | Design and deployment of a hybrid waste-to-energy system |                 | ✓          | ✓             |        | ✓                |
| Cardin et al. [54]                                    | Decision tree as part of four step flexibility analysis                     | Innovative nuclear reactor deployment                    |                 | ✓          | ✓             |        | ✓                |
| <b><i>Other sequential decision making papers</i></b> |                                                                             |                                                          |                 |            |               |        |                  |
| Chen et al. [37]                                      | Decision tree + Markov chain Monte Carlo                                    | Brugge reservoir pilot project                           |                 | ✓          | ✓             | ✓      | ✓                |
| Giuliani et al. [47]                                  | Markov decision processes + multi-objective evolutionary algorithm          | Red River reservoir system management                    | ✓               | ✓          | ✓             |        |                  |
| Wang and Rivera [49]                                  | Model predictive control                                                    | Semiconductor manufacturing supply chain management      |                 | ✓          | ✓             |        |                  |
| This paper                                            | Multi-objective decision tree + Monte Carlo simulation + Bayesian inference | lunar ISRU pilot and full-scale plant deployment         | ✓               | ✓          | ✓             | ✓      | ✓                |

model information-gathering. Table 1 lists the methods used in some of the past studies mentioned above in comparison with our proposed method. As can be seen in Table 1, the proposed method can address the key features of early-stage ISRU deployment.

The proposed method is demonstrated through a case study of deploying lunar ISRU pilot and full-scale plants in the southern polar region. Although past studies have designed lunar ISRU pilot plants [14, 39], a method to select an architecture for a pilot plant and its effect on the full-scale plant deployment has not been fully explored. This case study demonstrates the use of multi-objective decision tree analysis to examine the benefits and risks of each pilot plant architecture.

### 3. Methodology

#### 3.1. Overview

Fig. 1 shows the flowchart of the overall methodology applied in this paper. The shaded gray area represents the method previously employed by Ikeya et al. [20]. As depicted, this paper integrates Monte Carlo simulations of ISRU architectures, with multi-objective decision tree analysis (Section 3.2) and Bayesian inference (Section 3.3). Combinations of decisions selected through decision analysis are further examined by cumulative frequency analysis (Section 3.4).

#### 3.2. Multi-objective decision tree analyses

As explained in Section 2.2, the deployment of ISRU plants can be modeled as a multi-objective sequential decision-making problem [22]. Among the various methods available to solve such problems, this paper employs decision tree analysis for its relative simplicity and clarity in understanding and communicating key properties of the problem. It is also chosen due to the powerful insights it can provide, as shown in this study, prior to more sophisticated modeling and analysis.

Consider, for instance, a decision tree  $\mathcal{T}$  (see Fig. 2 for a sample two-stage decision tree), which consists of decision nodes  $D \in \mathcal{D}$ , chance nodes  $C \in \mathcal{C}$ , and terminal nodes  $T \in \mathcal{T}$ . There are also action branches  $a \in \mathcal{A}$ , and chance branches  $e \in \mathcal{E}$  connecting each node. A subscript

on each branch represents an index at that stage, and a superscript represents the path from the most upstream node to that branch. For instance, the  $k$ th action after an action  $a_i$  and a state of nature  $e_j^i$  is denoted by  $a_k^{i,j}$ . Similarly, a subscript of a node is an index number of an upstream branch, and a superscript represents a specific path from the first decision node to the current node.

Let  $r(\cdot)$  be a performance measure (i.e., typically a reward, but can also be a cost) of a tree path up to one node. For a multi-objective decision tree analysis, this value is a vector  $\mathbf{r} = [r_1, r_2, \dots, r_l]^T$  instead of a scalar as in a single-objective decision analysis. Each element in this vector  $r_1, r_2, \dots, r_l$  represents a different performance measure (e.g., system mass and required power in an ISRU example). We further define  $\mathcal{L}$  as a set of integers in a range  $[1, l]$ , i.e.,  $\mathcal{L} = 1, 2, \dots, l$ .

Both single- and multi-objective decision trees can be solved by backward induction, also known as the rollback, or “averaging out and folding back” method. This method determines the best decisions by estimating their performance measures and eliminating decisions expected to perform poorly, from the last chance nodes to the first decision node [22,60]. For a single decision problem, the optimality of a derived policy was discussed by Bellman [61]. Bellman’s decomposition was further discussed for multi-objective problems by Frini et al. [62] Algorithm 1 summarizes backward induction to solve a multi-objective decision tree proposed by Haimas et al. [22]

After drawing a tree representing the studied case (line 1), the performance measure vectors  $\mathbf{r}$  need to be mapped to all terminal nodes (line 2). Let the performance of one scenario following one tree path be

$$\mathbf{r} = f(\mathbf{x}, \mathbf{A}, \mathbf{E}), \quad (1)$$

where  $\mathbf{x}$  represents deterministic input parameters for the system. The other inputs to function  $f(\cdot)$ ,  $\mathbf{A}$  and  $\mathbf{E}$ , are actions taken and states of the system responding to the actions; therefore, they are sets of action branches  $a$  and chance branches  $e$  along one tree path. For example, consider a terminal node  $T_{\delta}^{\alpha,\beta,\gamma}$  of a two-stage decision tree after a path consisting of  $a_{\alpha}$ ,  $e_{\beta}^{\alpha}$ ,  $a_{\gamma}^{\alpha,\beta}$ , and  $e_{\delta}^{\alpha,\beta,\gamma}$ . The performance measure vector of a terminal node  $T_{\delta}^{\alpha,\beta,\gamma}$ ,  $\mathbf{r}(T_{\delta}^{\alpha,\beta,\gamma})$  is

$$\mathbf{r}(T_{\delta}^{\alpha,\beta,\gamma}) = f\left(\mathbf{x}, \left\{a_{\alpha}, a_{\gamma}^{\alpha,\beta}\right\}, \left\{e_{\beta}^{\alpha}, e_{\delta}^{\alpha,\beta,\gamma}\right\}\right). \quad (2)$$

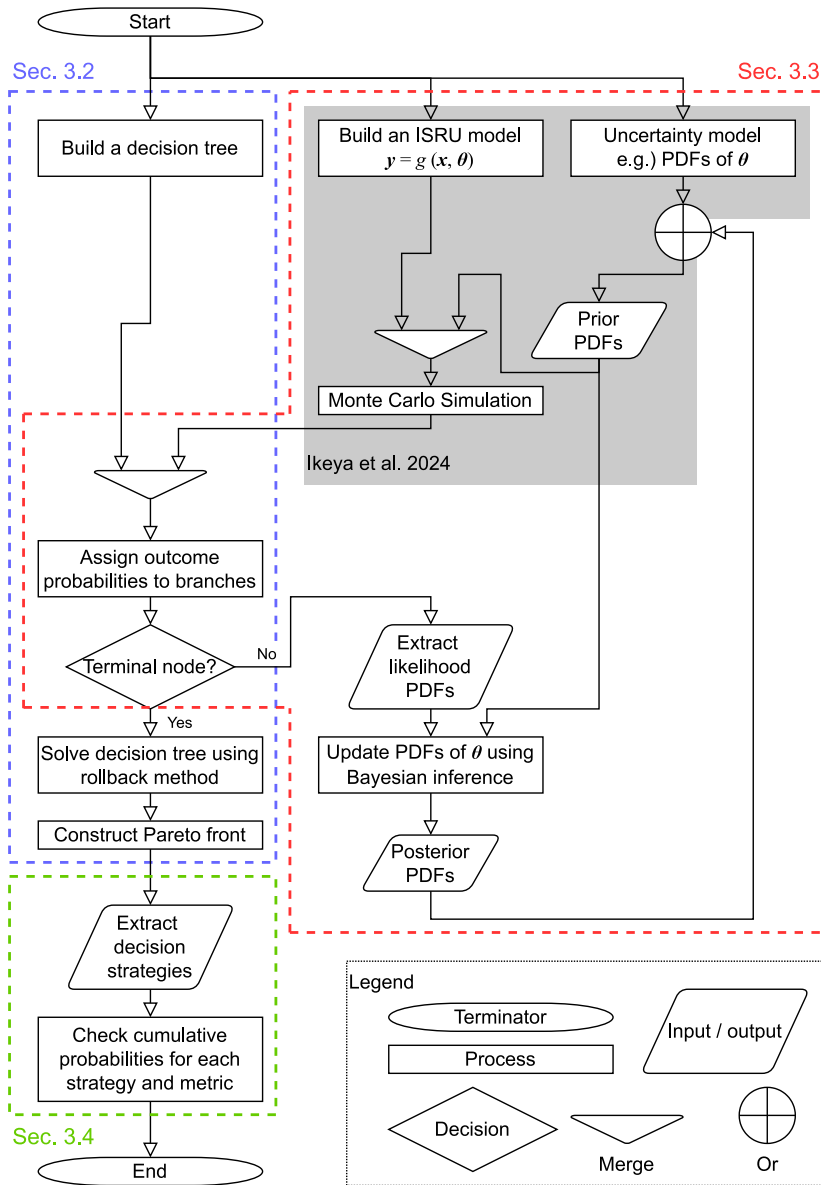


Fig. 1. Flowchart of the proposed methodology.

**Algorithm 1** Multi-objective decision tree analysis backward induction

```

1: Draw a decision tree  $\mathcal{T}$ 
2: Assign performance measure vectors  $\mathbf{r}$  to all terminal nodes  $T$ 
3: Assign an a priori probabilities or compute them (Section 3.3)
4: for each stage from the last to the first do
5:   if the last stage then
6:     Compute  $\mathbf{r}(C)$  (Eq. (3))
7:   else
8:     for  $\forall a$  in the corresponding Pareto set  $\mathcal{P}$  do
9:       Compute  $\mathbf{r}(C)$  (Eq. (5))
10:    end for
11:  end if
12:  Compute  $\mathbf{r}(D)$  (Eq. (8))
13:  Update Pareto set  $\mathcal{P}$ 
14: end for

```

Probabilities also need to be assigned to all chance branches (line 3). These probabilities  $p(e)$  can be predefined assuming uniform distribution in the states of the system [54] when further estimation is unavailable. Instead, this paper uses Monte Carlo simulations and Bayesian inference to assign these values (see Section 3.3 for a more detailed approach).

The backward induction starts with the calculation of the expected performance of a path up to the last chance nodes (line 6). See Fig. 3 for a sample chance node connected to the terminal nodes. Let the number of chance branches from each node be  $k$ . The performance measure vector  $\mathbf{r}$  of each decision can be calculated by averaging out all possible outcomes, i.e., downstream nodes, using preassigned probabilities. Consider a chance node  $C_{\gamma}^{\alpha,\beta}$  with  $k$  branches connected to terminal nodes  $\mathcal{T}^{\alpha,\beta,\gamma} = \{T_1^{\alpha,\beta,\gamma}, \dots, T_k^{\alpha,\beta,\gamma}\}$  (Fig. 3). A set of chance branches from this node is  $\mathcal{E}^{\alpha,\beta,\gamma} = \{e_1^{\alpha,\beta,\gamma}, \dots, e_k^{\alpha,\beta,\gamma}\}$ . We further define a set of pairs of a chance branch and its corresponding terminal node as  $(\mathcal{E}, \mathcal{T})^{\alpha,\beta,\gamma} = \{(e_1^{\alpha,\beta,\gamma}, T_1^{\alpha,\beta,\gamma}), \dots, (e_k^{\alpha,\beta,\gamma}, T_k^{\alpha,\beta,\gamma})\}$ . Then, the performance

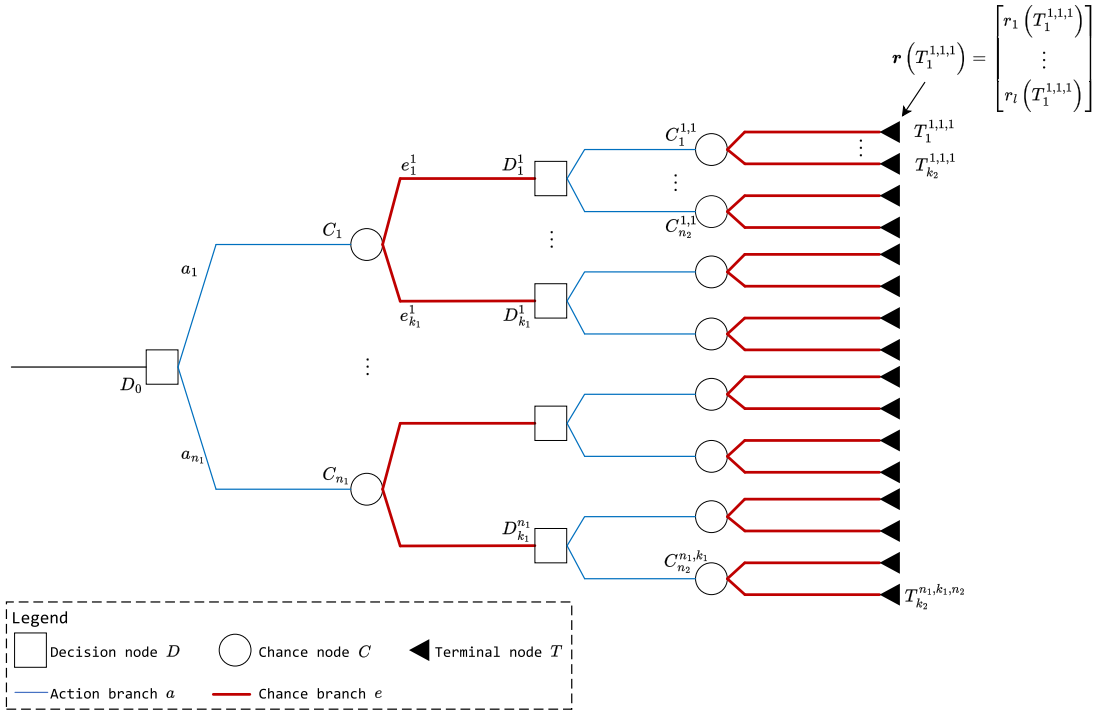


Fig. 2. An example of a two-stage decision tree.

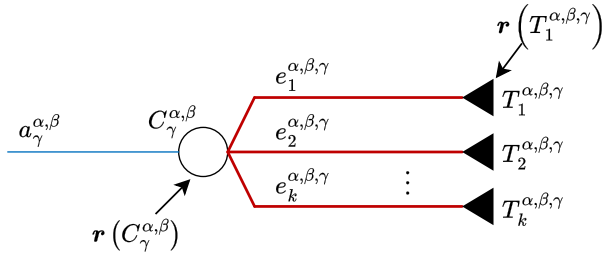


Fig. 3. One of the most downstream chance nodes connected to terminal nodes.

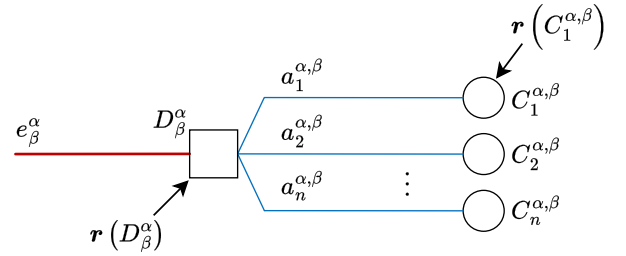


Fig. 4. One decision node connected to chance nodes.

measure vector of  $C_\gamma^{\alpha,\beta}$  can be expressed as

$$r(C_\gamma^{\alpha,\beta}) = \sum_{(e,T) \in (\mathcal{E}, \mathcal{T})^{\alpha,\beta,\gamma}} p(e | a_\gamma^{\alpha,\beta}) r(T), \quad (3)$$

where  $p(e | a_\gamma^{\alpha,\beta})$  expresses the probability of a state  $e$  occurring under the previous action  $a_\gamma^{\alpha,\beta}$ . The summation of probabilities of all possible states of nature is 1, i.e.,

$$\sum_{e \in \mathcal{E}^{\alpha,\beta,\gamma}} p(e | a_\gamma^{\alpha,\beta}) = 1. \quad (4)$$

Similarly, the calculation of the performance measure of upper-stream chance node  $C_\alpha$  connected to chance branches and decision nodes  $(\mathcal{E}, D)^\alpha = \{(e_1^\alpha, D_1^\alpha) \dots, (e_k^\alpha, D_k^\alpha)\}$  (line 9) is done by

$$r(C_\alpha) = \sum_{(e,D) \in (\mathcal{E}, D)^\alpha} p(e | a_\alpha) r(D). \quad (5)$$

After computing the performance measure vectors of chance nodes (Eq. (3)), we focus on upper-stream decision nodes connected to these chance nodes. At a decision node, all performance measure vectors of downstream chance nodes are compared (line 12) to generate a Pareto set (line 13).

A performance measure vector  $r(C_\gamma)$  is said to be non-dominated if there does not exist a performance measure vector  $r(C_{\gamma'})$  such that

$$\forall \lambda \in \mathcal{L}, r_\lambda(C_\gamma) \leq r_\lambda(C_{\gamma'}), \quad (6)$$

$$\exists \lambda \in \mathcal{L}, r_\lambda(C_\gamma) < r_\lambda(C_{\gamma'}). \quad (7)$$

The actions corresponding to the non-dominated performance measures are saved into a Pareto set  $\mathcal{P}$ . Consider a decision node  $D_\beta^\alpha$  connected to  $n$  action branches  $\mathcal{A}^{\alpha,\beta}$  (Fig. 4). Unlike single-objective decision tree analysis, the performance measure at this chance node becomes a set of non-dominated vectors:

$$r(D_\beta^\alpha) = \{r(C_n^{\alpha,\beta}) | a_n^{\alpha,\beta} \in \mathcal{P}^{\alpha,\beta}\}, \quad (8)$$

where  $\mathcal{P}^{\alpha,\beta}$  is a Pareto optimal set of downstream actions connected to  $D_\beta^\alpha$ .

Fig. 5 depicts a Pareto front in the case of three actions and two objectives. In this example figure, action  $a_3^{\alpha,\beta}$  can be discarded for further analysis, and  $\mathcal{P}^{\alpha,\beta} = \{a_1^{\alpha,\beta}, a_2^{\alpha,\beta}\}$ .

This process of calculating performance measure vectors from the most downstream chance nodes to upstream nodes is repeated until a Pareto set of the most upstream decision nodes is obtained (lines 4–13). The obtained Pareto set can visualize the trade space between the studied metrics. Although the introduced method here is used to maximize the value of each performance measure, changing the inequality signs in Eqs. (6) and (7) can yield combinations of decisions to minimize each performance measure.

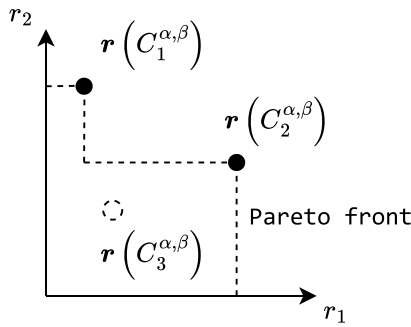


Fig. 5. Non-dominated performance measures and a dominated performance measure.

### 3.3. Monte Carlo simulation and Bayesian inference

In this paper, the probabilities assigned to the decision tree are estimated from Monte Carlo simulations. This method generates randomized input parameters from predefined probability density functions. Monte Carlo simulations generate distributions of possible system's performance, subject to stochastic variations. Monte Carlo simulations have been previously used to estimate the performance of ISRU plants [20,21].

Fig. 6 depicts a more detailed flowchart of how Monte Carlo simulations, Bayesian inference, and a decision tree are integrated. This corresponds to the area bounded by the red dashed line in Fig. 1. Unlike the backward induction explained in Section 3.2, the pre-assignment of probabilities is conducted from the first stage to the last stage of the decision tree.

First, a model that maps the input parameters to output performance measures of a system under study is created:

$$y = g(x, \theta), \quad (9)$$

where  $x$  and  $\theta$  are the deterministic and uncertain input parameters (i.e., Monte Carlo parameters), respectively. The randomized parameters  $\theta = [\theta_1, \theta_2, \dots, \theta_j]^T$  depend on prior distributions of each element of  $\theta$ . We assume that the probability distributions of these Monte Carlo parameters can be estimated for the first stage. For instance, Cilliers et al. [21] and Ikeya et al. [20] estimated distributions of uncertain parameters such as resource content from past studies.

Given model  $g(x, \theta)$  and the distributions of  $\theta$ , the Monte Carlo method outputs histograms of each performance measure. By adjusting the width and number of bins, the probability of each chance branch can be obtained.

For multi-stage decision analysis,  $\theta$  at the later stages of the tree depends on the tree path,  $(A, E)$ , as the information obtained varies depending on the actions taken and the state of nature after each action. Bayesian inference is utilized to update the probabilities of  $\theta$  in this paper.

Let  $\Theta$  be a set of all generated combinations of  $\theta$ . Assuming the system state metric used for each chance node is derived as the  $i$ th element of the function  $g(\cdot)$ :

$$s = g_i(x, \theta). \quad (10)$$

Then, a subset of  $\Theta$  that leads to a chance branch  $e$ ,  $\Theta_e$ , can be represented as:

$$\Theta_e = \{\theta | g_i(x, \theta) \approx e\}. \quad (11)$$

From the extracted subset  $\Theta_e$  for each chance branch, likelihood distributions for each element of  $\theta$  can be approximated. Using the prior distribution and the approximated likelihood distribution, the posterior distribution can be calculated based on Bayesian inference. For instance, for an element of  $\theta$ ,  $\theta$ , given prior probability  $p(\theta)$

and an approximation of likelihood probability  $p(\Theta_e | \theta)$ , the posterior probability becomes

$$p(\theta | \Theta_e) = \frac{p(\Theta_e | \theta)p(\theta)}{p(\Theta_e)}. \quad (12)$$

The obtained posterior probability distributions are then used as prior distributions for the next stage of Monte Carlo simulations, continuing this process until reaching the terminal nodes.

### 3.4. Evaluation of decisions under uncertainty

The backward induction method introduced in Section 3.2 constructs a Pareto set given multiple metrics based on the expected value of the system following each decision strategy. This approach, however, may overlook potential risks and benefits when computing expected values. To thoroughly evaluate each decision strategy in a Pareto set, computing the cumulative probabilities (also known as target curves) of each metric for one decision strategy is essential. This can be done by estimating the possible outcomes of performance measures and their corresponding probabilities following tree paths [42,54].

Finally, the expected value of flexible operations is estimated inspired by the VoI and VoF approaches. Comparisons of the target curves of each decision strategy on the produced Pareto front for each metric can estimate this value.

## 4. Case study

### 4.1. Overview and assumptions

This paper examines the early-stage deployment of lunar ISRU plants in the lunar southern polar region considering uncertainty in resource and operational availability and technology efficiency. The southern polar region has attracted significant interest for Artemis Base Camp [1] for two main reasons. First, certain areas in this region are known for their near continuous sunlight availability [63]. Second, the presence of water ice has been experimentally confirmed by the Lunar CRater Observation and Sensing Satellite (LCROSS) [64].

This case study focuses solely on oxygen as a product of ISRU. While other products, such as hydrogen and metals, are valuable for human exploration, their inclusion would unnecessarily complicate the analysis and, therefore, is beyond the scope. Note that the same method explained in Section 3 can be used for other products as long as there is a model that can output objectives of interest and state of nature, and take uncertain parameters (Eq. (9)). See Section 5.2 for further discussion.

Fig. 7 depicts a simple version of the decision tree for this case study. The tree consists of two stages, a pilot plant and full-scale plant deployment. Due to limited knowledge of the lunar environment and ISRU operations, systems engineers and mission architects must plan ISRU plant projects under uncertainty. For terrestrial reservoir management [37] and industrial plants [65], operating pilot projects is crucial for gathering more information and reducing risks for future full-scale operations. Indeed, the deployment of a lunar ISRU pilot plant prior to a full-scale plant has also been suggested [14,38,66].

Table 2 outlines the considered decisions and states of nature in this paper. The first decision node corresponds to the oxygen extraction architecture selection for a pilot-scale ISRU plant. We consider four options: Carbothermal Reduction (CR) of dry regolith, direct Water Extraction (WE) from icy regolith, and two hybrid architectures that integrate these technologies: Parallel Hybrid (PH) and Series Hybrid (SH) architectures [20]. Each architecture is explained in detail in Section 4.2.

In this paper, the states of nature  $s$  (Eq. (10)) is defined by the relative oxygen yield compared to the expected yield. While making the states of both stages continuous is feasible, it would be too computationally expensive. Thus, we define the first-stage chance nodes with

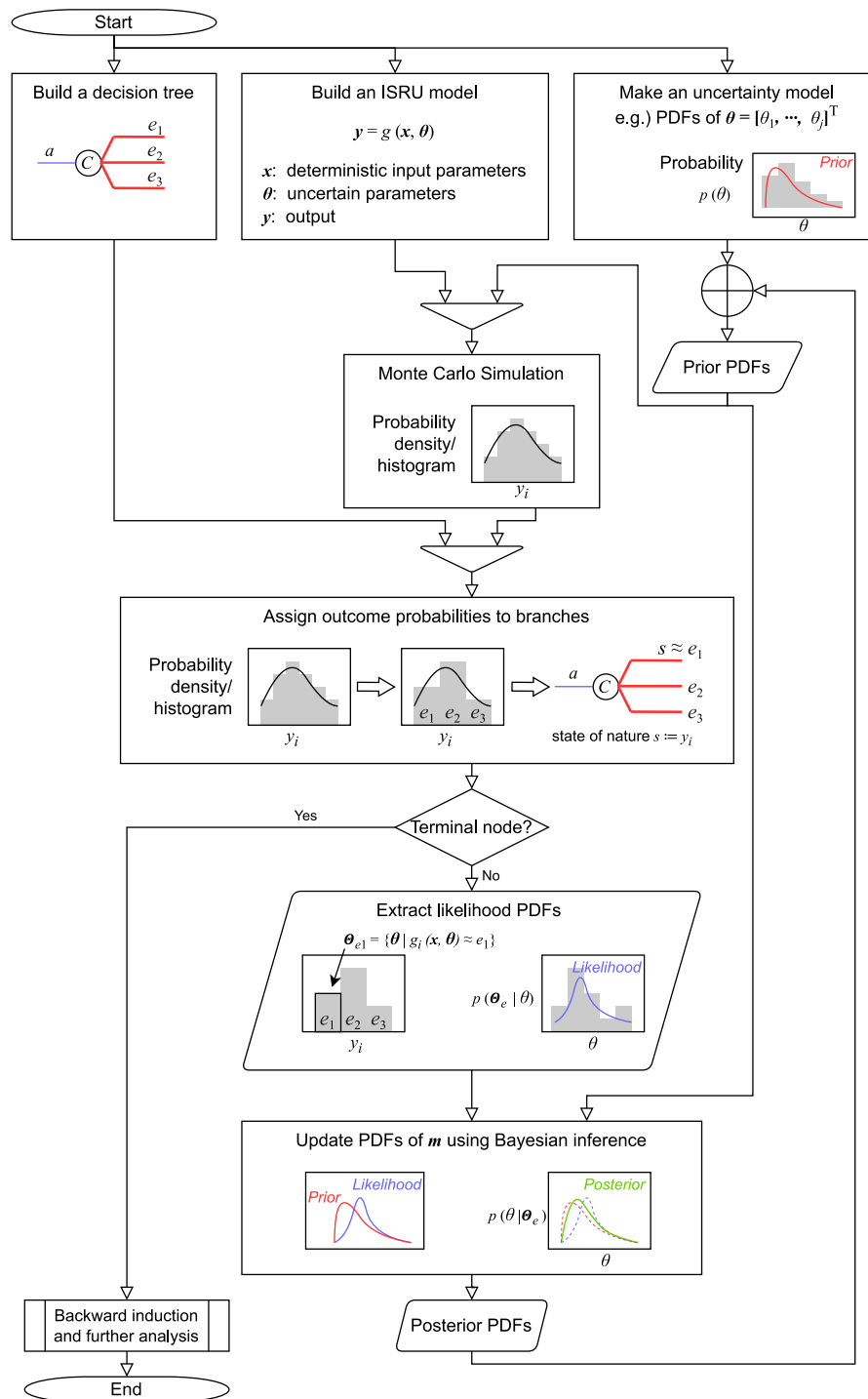


Fig. 6. Flowchart of the Monte Carlo simulation and Bayesian uncertainty update.

discrete branches and the second-stage nodes with a continuous state of nature. In the first stage, the relative oxygen yield is discretized into three cases for dry regolith and icy regolith: 0%–90%, 90%–110%, and 110% or higher. These numbers for discretization are selected to avoid extremely low probability for any cases in the first stage. The impact of using discrete branches in the first stage should be assessed in future work.

Based on the relative oxygen yield in the initial deployment, a decision-maker can pick an architecture for a full-scale plant in the subsequent stage. Historically, the deployment of new space systems is developed and proven in a step by step manner to reduce risk which

generally involves deploying subscale development systems prior to committing to larger systems. Therefore, we assume that any technology not tested at the pilot scale will not be deployed at the full scale. Additionally, we consider potential modifications in the PH architecture, allowing adjustments to the production ratio between carbothermal reduction and water extraction.

#### 4.2. Oxygen extraction architectures

This paper adopts models, assumptions, and ConOps outlined by Ikeya et al. [20] All architectures employ the same surface excavator

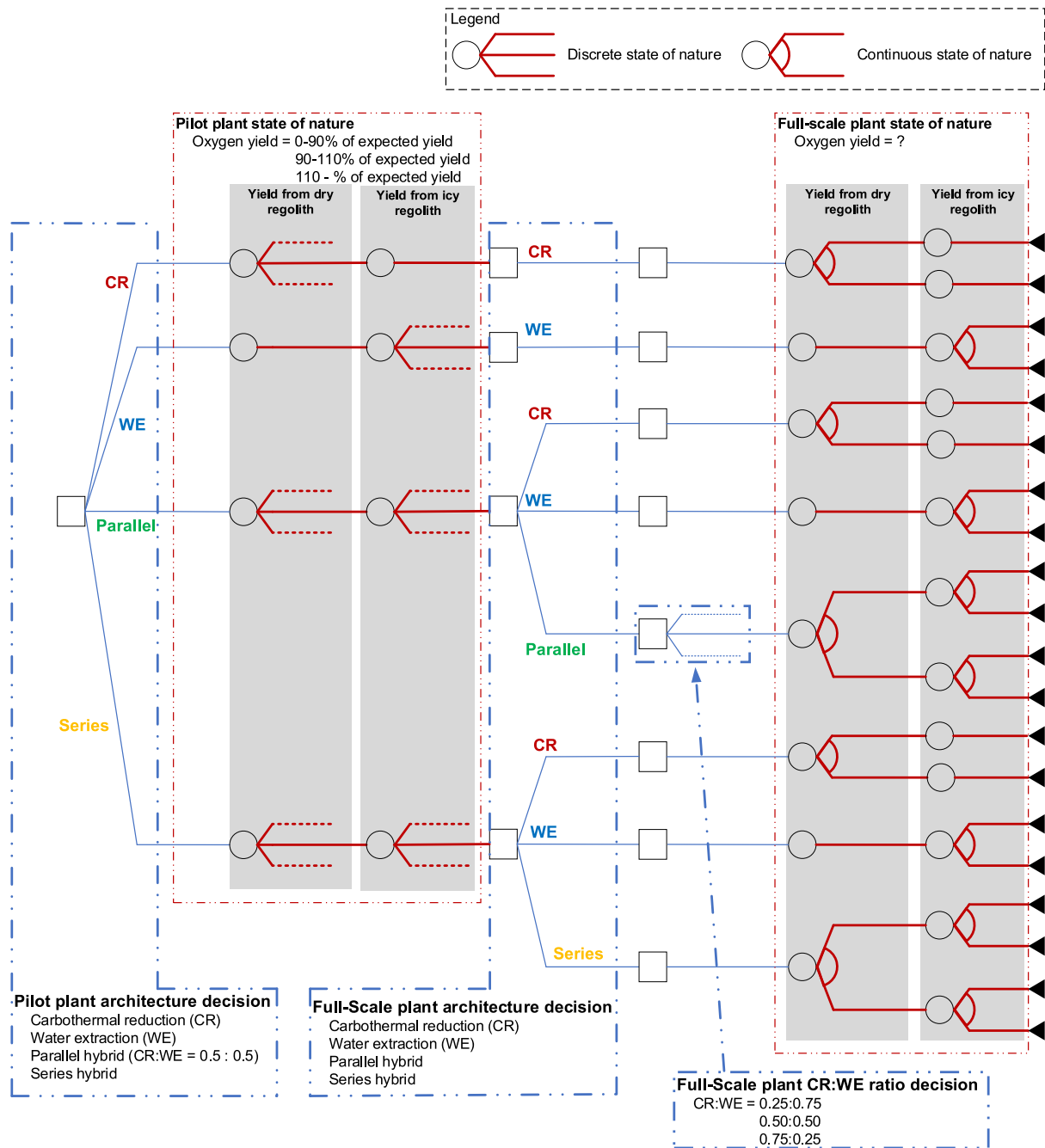


Fig. 7. Decision tree of for the case study.

RASSOR 2.0 [67], Solid Oxide Electrolysis (SOXE), liquefaction, and storage subsystems. Additionally, this paper does not consider hydrogen liquefaction or storage and only focuses on oxygen production. The interested readers are referred to in-depth explanations of each architecture by Ikeya et al.

#### 4.2.1. Carbothermal reduction

In general, ISRU from planetary regolith involves three phases: mining, beneficiation, and extraction [32]. In the mining phase, excavators are used to dig the planetary surface and collect the regolith. These excavators transfer the collected regolith to the next destination where the beneficiation occurs. Beneficiation is a process of preparing for the chemical reaction by sorting the particle size of the regolith and enriching the mineral of interest to increase the efficiency of the chemical reaction. After the beneficiation, the beneficiated regolith is

reduced to a target product following the thermo- and electrochemical reaction.

CR followed by methanation can be expressed as:



where  $\text{MeO}_x$  and  $\text{Me}$  represent metal oxide and metal, respectively. CR can process pyroxene, ilmenite, and olivine primarily [68]. Fig. 8 depicts the simplified CR process. In this paper, following the assumption made by Ikeya et al. [20], we consider for simplicity only the reaction between carbon and silica ( $\text{SiO}_2$ ). Depending on the temperature [12, 69,70], pressure [70], and C/ $\text{SiO}_2$  mole ratio [69], the final equilibrium compositions of the CR vary. In this paper, we simply assume a reaction

**Table 2**  
Decisions and chances implemented in the case study tree.

| Type                                                | Branches                | Note                                            |
|-----------------------------------------------------|-------------------------|-------------------------------------------------|
| <i>First stage decision</i>                         |                         |                                                 |
| Architecture                                        | CR                      | O <sub>2</sub> production ratio CR:WE = 0.5:0.5 |
|                                                     | WE                      |                                                 |
|                                                     | Parallel hybrid CR + WE |                                                 |
|                                                     | Series hybrid CR + WE   |                                                 |
| <i>First stage chance</i>                           |                         |                                                 |
| Relative O <sub>2</sub> yield from dry/icy regolith | Good                    | 110% or more of expected yield                  |
|                                                     | Normal                  | 90%–110% of expected yield                      |
|                                                     | Bad                     | 0%–90% of expected yield                        |
| <i>Second stage decision</i>                        |                         |                                                 |
| Architecture                                        | Same as pilot           | Hybrid pilot plant case only                    |
|                                                     | Reduce to CR            |                                                 |
|                                                     | Reduce to WE            |                                                 |
| Design                                              | CR:WE = 0.25:0.75       | Parallel hybrid full-scale plant case only      |
|                                                     | CR:WE = 0.5:0.5         | Parallel hybrid full-scale plant case only      |
|                                                     | CR:WE = 0.75:0.25       | Parallel hybrid full-scale plant case only      |
|                                                     |                         |                                                 |
| <i>Second stage chance</i>                          |                         |                                                 |
| Relative O <sub>2</sub> yield from dry/icy regolith | N/A                     | Continuous state of nature                      |

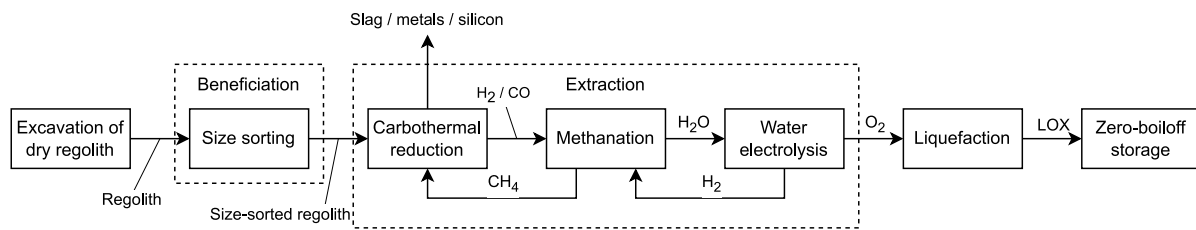


Fig. 8. Carbothermal reduction of dry regolith.

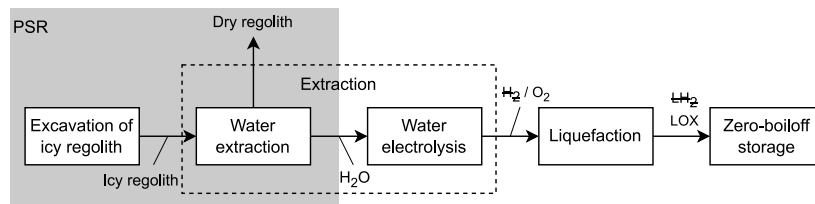


Fig. 9. Water extraction from excavated icy regolith. Although this process produces hydrogen, its liquefaction and storage is not considered in this paper.

intermediate SiC is fully reduced and the production of SiO is ignored. The following represents the overall reaction of SiO<sub>2</sub> corresponding to Eq. (14):



For details of carbothermal reduction of silica, please refer to the studies by Troisi et al. [12] and Lee et al. [69].

#### 4.2.2. Water extraction from icy regolith

Since LCROSS experimentally confirmed water ice on the Moon [64], various water extraction approaches from icy regolith in permanently shadowed regions (PSRs) have been proposed. Kiewiet et al. [30] compared three different water extraction methods: heating excavated regolith, heating the icy regolith surface in a heat dome, and using rods placed deep in the ground in a heat dome. Their results indicate the potential performance improvement of the excavation method; however, they also noted that the complexity of the ConOps could pose more challenges. The flowchart of this process is shown in Fig. 9. Although the flowchart may appear simple, a lack of solar power in mining sites and additional water tankers that travel between a PSR and a sunlit area complicate this approach.

#### 4.2.3. Hybrid lunar ISRU architectures

Ikeya et al. have proposed two hybrid ISRU architectures [20]. In the Parallel Hybrid (PH) architecture, there are two separate mining

sites. At a Peak of Eternal Light (PEL), dry regolith undergoes the carbothermal reduction process. Meanwhile, at a PSR, icy regolith is processed to extract water. The extracted water is transported from the PSR to the PEL using water tankers. Once at the PEL, the transported water, along with the water produced from the carbothermal reduction, is electrolyzed, liquefied, and stored. Fig. 10 summarizes the general process of oxygen production in this architecture. The ratio of oxygen produced from dry regolith to icy regolith can be adjusted by appropriately sizing each processing plant, which is used as one of the decisions to make for the full-scale plant in this case study.

The other architecture proposed by Ikeya et al. [20] connects the water extraction and the carbothermal reduction process in series. In this architecture, the remaining dry regolith after water extraction is further processed through carbothermal reduction in PSR (Fig. 11). While this architecture may lead to the largest mass, it could require a relatively slower excavation rate.

A specific mining site in the lunar south pole is not identified due to lack of information. For WE and the hybrid architectures, we assume that the traverse distance for water tankers between a PEL and a PSR is 5,000 m, based on previous research [13,71,72].

#### 4.3. Metrics for in-situ resource utilization

Traditionally, past ISRU system studies have evaluated system mass [18,20,73], required power [18,20], and mass payback ratio [18] as

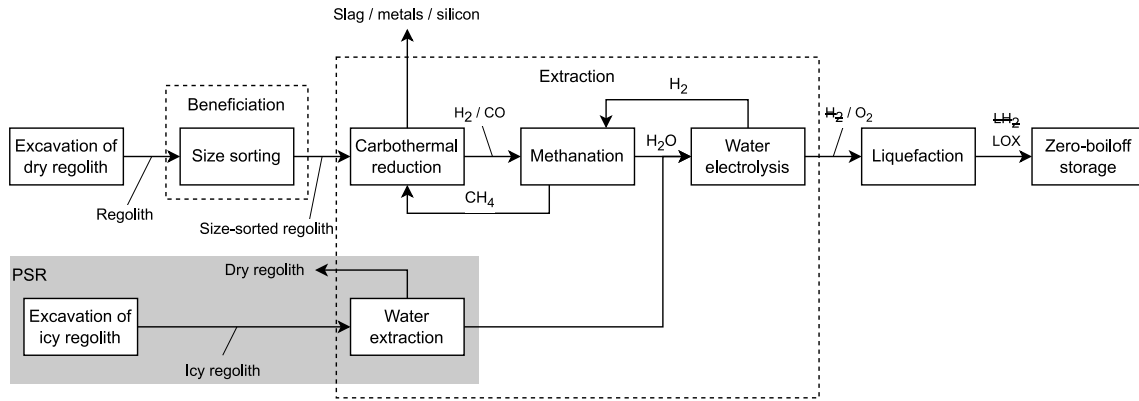


Fig. 10. Oxygen extraction process in the parallel hybrid ISRU architecture.

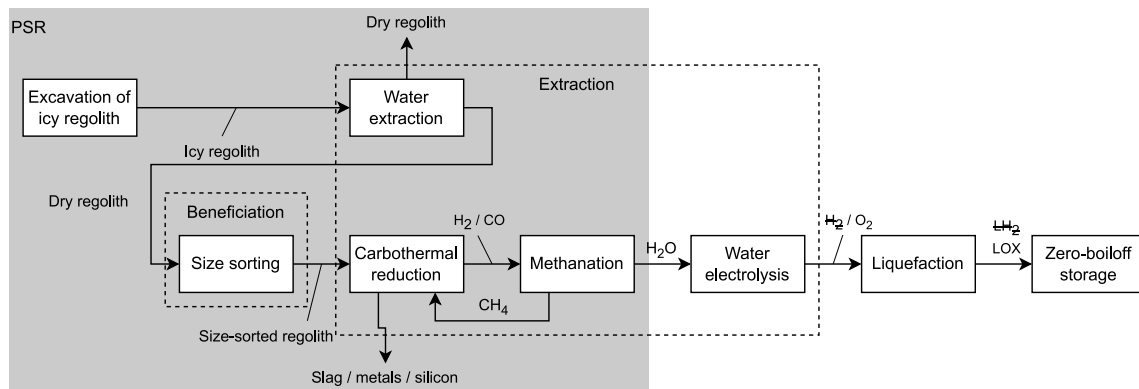


Fig. 11. Oxygen extraction process in the series hybrid ISRU architecture.

performance measures. The mass payback ratio is defined as the system mass required to produce a unit mass of a product. Product (e.g., oxygen) mass, product yield, and excavation rate are also often used as performance indicator for plant design [20,21]. For Mars ISRU, Hinterman et al. [74] conducted a multi-objective optimization of the system employing mass, power, volume, risk, and cost. Similarly, this study considers evaluating ISRU plants using multiple performance metrics, rather than just one, to provide a more comprehensive assessment for future decision-making.

The objectives considered in this paper are mass payback ratio, oxygen yield, and the mass of produced oxygen per unit input energy. The mass payback ratio can work as a good indicator of the cost of the plant. Note that when the mass payback is smaller than 1.0, deploying ISRU might be less appealing than directly importing oxygen to the Moon from Earth (ignoring the LOX tank mass). Oxygen yield measures how much oxygen a plant produces given a unit mass of excavated regolith, so this value is always smaller than 1.0. To preserve the scientific value of lunar regolith and minimize environmental impact, designs requiring extensive regolith excavation are less favorable. The mass-energy efficiency is another crucial factor in ISRU design. Sunlight availability on the Moon is often limited in areas rich in resources like water ice, making energy-intensive systems technologically challenging or even infeasible. For all of these three metrics, the larger the value, the better the performance is, i.e., maximizing the expected values of these is the objective in this case study.

#### 4.4. Fixed and varying parameters

Table 3 lists fixed parameters for the case study. Pilot plants were scaled to produce 1 t of oxygen over a year. For full-scale plants, as a baseline case, 10 t of oxygen production per year and 10 years of

Table 3

Fixed parameters for the case study.

| Parameters                         | Assumed values          |
|------------------------------------|-------------------------|
| <b>Pilot plant parameters</b>      |                         |
| Operational period                 | 1 year                  |
| Target production                  | 1 t of oxygen           |
| <b>Full-scale plant parameters</b> |                         |
| Operational period                 | 10 years                |
| Target production                  | 10 t of oxygen per year |
| <b>Common parameters</b>           |                         |
| PEL/PSR traverse distance          | 5000 m                  |
| Mass margin                        | 30%                     |
| Power margin                       | 30%                     |
| PEL avg. temperature               | 200 K                   |
| PSR avg. temperature               | 80 K                    |
| Outer space temperature            | 4 K                     |

operation are selected. The effects of the scale of a full-scale plant and its operational period are discussed through sensitivity analysis (see Section 4.6).

As mentioned in Section 3.3, the proposed method employs Monte Carlo simulations to assign probabilities to each chance branch. Table 4 summarizes the Monte Carlo parameters used for the first decision branches of this case study. These distributions are updated depending on the first chance branches using Bayesian inference (Section 3.3). For each decision branch, Monte Carlo simulations with 5000 scenarios, i.e., 5000 different combinations of parameters listed in Table 4 are computed.

Table 5 lists the average mass and required total energy of each plant architecture. The landed mass is estimated using a model developed by Ikeya et al. [20], as well as models in ISRUlib [79], an open-source python based database of ISRU subsystems. Each landed

**Table 4**

Monte Carlo parameters for uncertainty consideration for the first stage. See Ikeya et al. [20] for a more detailed explanation of each parameter.

| Variable                             | Distribution ( $\mu, \sigma^2$ )  | Reference |
|--------------------------------------|-----------------------------------|-----------|
| Particle size mass fraction [wt%]    | $\mathcal{N}(62.6, 8.3^2)$        | [75]      |
| Silica content [wt%]                 | $\mathcal{N}(45.2, 0.93^2)$       | [76]      |
| Water ice content [wt%]              | $\text{Lognormal}(\ln(4), 0.3^2)$ | [64,77]   |
| Water yield [–]                      | $\mathcal{N}(0.75, 0.05^2)$       | [13,78]   |
| Operational availability [days/year] | $\mathcal{N}(200, 10^2)$          | [63]      |

**Table 5**

Average landed system mass and required total energy of each plant design. Hydrogen liquefaction and its storage subsystems are excluded from the estimation.

| Architecture           | Pilot (1 t O <sub>2</sub> for 1 year) |                   | Full-scale (10 t O <sub>2</sub> per annum for 10 years) |                   |
|------------------------|---------------------------------------|-------------------|---------------------------------------------------------|-------------------|
|                        | Mass [t]                              | Energy [1e+4 kWh] | Mass [t]                                                | Energy [1e+6 kWh] |
| CR                     | 0.7                                   | 3.4               | 14.3                                                    | 3.5               |
| WE                     | 0.7                                   | 1.6               | 5.3                                                     | 1.3               |
| PH (CR:WE = 0.5:0.5)   | 1.0                                   | 2.5               | 13.8                                                    | 2.4               |
| PH (CR:WE = 0.75:0.25) | N/A                                   | N/A               | 18.3                                                    | 2.9               |
| PH (CR:WE = 0.25:0.75) | N/A                                   | N/A               | 9.6                                                     | 1.8               |
| SH                     | 1.0                                   | 3.0               | 28.3                                                    | 2.7               |

mass includes the mass of subsystems such as excavators, water tankers, beneficiation subsystems, reactors, SOXE subsystem, oxygen liquefaction, and zero-boiloff storage systems when they are required. Reactant mass for the CR process is also added to the landed mass assuming 91% recycle efficiency of both hydrogen and methane [80]. Hydrogen liquefaction and its storage subsystems are eliminated from the landed mass and extraction energy estimation since these subsystems unfairly biases the system mass and required energy of the WE, PH and SH architectures. As discussed by Ikeya et al. the landed mass varies depending on the confidence level in meeting the target production and some uncertain parameters. In this paper, all architectures are designed to meet the target production with 95% confidence by adjusting the number of excavators, reactors, and cryocoolers for liquefaction.

#### 4.5. Results from the baseline case

Fig. 12 shows Pareto optimal points of all combinations of decisions. Each axis represents one metric: oxygen yield ( $m_{\text{O}_2}/m_{\text{reg}}$ ), mass payback ( $m_{\text{O}_2}/m_{\text{sys}}$ ) and mass-energy efficiency ( $m_{\text{O}_2}/E_{\text{total}}$ ). Each point in the figure represents one decision strategy and each color represents different pilot plant architecture. Under our assumptions, we do not consider any decisions for full-scale plant deployment after deploying either a CR or WE pilot plant (Fig. 7). As a result, only one decision strategy (one point in Fig. 12) exists for each of the CR and WE pilot plant cases. Conversely, after deploying either a parallel or series hybrid pilot plant, we can select an architecture (as well as CR and WE dependence ratio for the PH full-scale plant) based on the relative oxygen yield from both dry and icy regolith. This flexibility in full-scale plant deployment results in multiple data points on the Pareto front. Some representative strategies are discussed further later.

Fig. 13 shows the two-dimensional projections of the Pareto front in Fig. 12. Comparing these plots reveals that the decision strategies on the 2D Pareto front vary depending on the selected objectives. For instance, in Fig. 13(a), choosing either CR, WE, PH, or SH pilot plant is on the 2D Pareto front. However, in Fig. 13(c), only the WE dedicated strategy appears on the 2D Pareto optimal set. This indicates the potential risk a decision-maker may overlook when considering only two objectives.

For further analysis, this study selects five decision strategies after deploying a parallel hybrid plant (P.0 – P.4) and five strategies for a series hybrid pilot plant (S.0 – S.4). Table 6 lists these strategies based on the yield from the pilot plant. For instance, strategy P.0 deploys a WE full-scale plant if the yield from dry regolith is less than 90% of the expected yield and more than 110% from icy regolith; otherwise, it deploys a CR full-scale plant. Strategy S.0 deploys a series hybrid

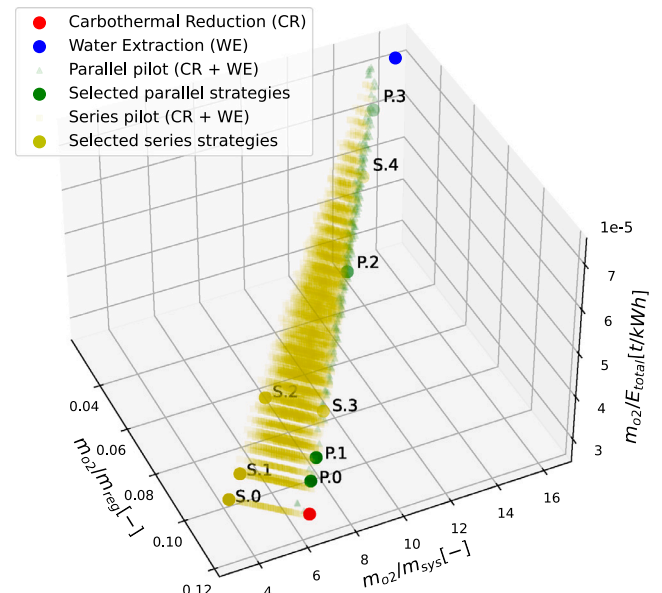


Fig. 12. Three-dimensional Pareto front for case study. Different pilot plant architectures are plotted in different colors. Each point represents one decision strategy. Parallel or series strategies are the strategies that deploy a PH or SH pilot plant, but not necessarily hybrid architecture for full-scale. Selected strategies are summarized in Table 6.

architecture at full scale regardless of the yield from the series pilot hybrid plant. These strategies are also plotted in Figs. 12 and 13.

Among all strategies on the 3D Pareto front that deploy a PH pilot plant, there are only a few of them that also include a PH full-scale plant deployment. Strategy P.4 is an example of them, which deploys a PH plant in full-scale only when the yield from dry regolith is good (more than 110% of the expected yield) and medium (between 90% and 110%) for icy regolith. Intuitively, this strategy of deploying a PH full-scale only under this condition but not when the relative dry regolith yield is good and that of icy regolith is bad is unnatural. This indicates that a decision-maker needs to assess strategies on the Pareto front carefully before actually implementing any of them.

To assess the potential benefits and risks of each strategy, cumulative probabilities are estimated using the method outlined in Section 3.4 (Fig. 14). As mentioned in Section 4.1, the state of nature

**Table 6**  
Selected hybrid strategies.

| Pilot architecture |        | Parallel hybrid |     |     |     |                           | Series hybrid |     |     |     |     |
|--------------------|--------|-----------------|-----|-----|-----|---------------------------|---------------|-----|-----|-----|-----|
| Yield from pilot   |        | P.0             | P.1 | P.2 | P.3 | P.4                       | S.0           | S.1 | S.2 | S.3 | S.4 |
| Dry                | Ice    |                 |     |     |     |                           |               |     |     |     |     |
| Bad                | Bad    | CR              | CR  | CR  | WE  | WE                        | SH            | SH  | SH  | CR  | WE  |
|                    | Medium | CR              | WE  | WE  | WE  | WE                        | SH            | SH  | SH  | CR  | WE  |
|                    | Good   | WE              | WE  | WE  | WE  | WE                        | SH            | WE  | WE  | WE  | WE  |
| Medium             | Bad    | CR              | CR  | WE  | WE  | WE                        | SH            | SH  | SH  | CR  | WE  |
|                    | Medium | CR              | CR  | CR  | WE  | WE                        | SH            | SH  | SH  | CR  | WE  |
|                    | Good   | CR              | CR  | WE  | WE  | WE                        | SH            | SH  | WE  | WE  | WE  |
| Good               | Bad    | CR              | CR  | CR  | CR  | WE                        | SH            | SH  | SH  | CR  | CR  |
|                    | Medium | CR              | CR  | CR  | WE  | PH<br>(CR:WE = 0.75:0.25) | SH            | SH  | SH  | CR  | CR  |
|                    | Good   | CR              | CR  | CR  | WE  | WE                        | SH            | SH  | SH  | CR  | CR  |

in the first stage is discrete whereas that in the second stage is continuous. This results in a multi-modal distribution for each performance indicator, as seen in strategies P.0-P.4 and S.0-S.4 in Fig. 14.

For oxygen yield (Fig. 14(a)), deploying the CR architecture in both stages appears to be the most effective strategy overall, with strategy S.0 showing nearly equivalent performance. However, this strategy of deploying CR architecture in both stages performs the worst in terms of mass-energy efficiency (Fig. 14(c)). On the other hand, deploying the WE architecture in both stages can be the best option regarding both mass payback (Fig. 14(b)) and mass-energy efficiency (Fig. 14(c)). Fig. 14(b) also shows clear disadvantage of S.0, S.1, and S.2 strategies regarding mass payback. This is because of the significantly heavy landed mass for SH full-scale plant (see Table 5).

Table 7 summarizes the values of each metric for each decision strategy, showing the 5th and 95th cumulative probabilities (or percentiles), as well as the expected value. In this case study, the VoI is defined as the difference between the best hybrid strategy (P.0-P.4 or S.0-S.4) and the best conventional strategy (CR or WE). For example, in terms of oxygen yield (Table 7), the expected VoI is −0.2% comparing the CR strategy with the S.0 strategy. This negative sign indicates that the conventional strategy is advantageous compared to the hybrid strategy regarding oxygen yield. However, as can be seen in Fig. 14 and Table 7, many hybrid strategies also do not perform the worst while the CR or WE can be the worst-performing strategy in one metric. In other words, while the CR or WE strategy can be the best depending on the choice of metric, hybrid strategies are a good trade-off across all metrics, and hence, represent more balanced strategies.

#### 4.6. Sensitivity analysis

##### 4.6.1. Production period and scale

To see the effect of the scale and the operational period of a full-scale plant, changes in the two-dimensional Pareto fronts are assessed with different assumptions. We consider 5, 10, and 15 tons of oxygen per year as scales of full-scale plant, and 5, 10, and 15 years of its operation. Fig. 15 visualizes two-dimensional Pareto optimal strategies under each assumption.

Fig. 15(a) shows a significant shift in the Pareto front as the operational period changes. For the WE strategy, the mass payback ( $m_{O_2}/m_{sys}$ ) and operational year have a linear relationship. This is because  $m_{O_2}$  a product of produced oxygen mass per year and operational period, and the denominator, the landed system mass  $m_{sys}$  is a constant value. In contrast, for the other architectures, the landed system increases as the operational period increases due to the imperfect recycling of reactants for the CR process. A comparison of the same production period but a different scale suggests producing a larger amount of oxygen is generally more advantageous than smaller-scaled production in terms of the mass payback ratio.

Fig. 15(b) indicates that the production scale has a more substantial impact on the Pareto front between oxygen yield ( $m_{O_2}/m_{reg}$ ) and energy efficiency ( $m_{O_2}/E_{total}$ ) than the operational period does. This is because

both of these metrics do not depend on the operational period, whereas the scale of the plant indeed affects the required energy. Furthermore, from these Pareto fronts, producing 5 tons of LOX seems more efficient than 15 tons of LOX regarding energy.

From the discussions above, the result suggests that the larger-scale production designs are better in terms of mass payback while worse when considering mass-energy efficiency. One main reason for this trend is that the CR reactors and water extractors are designed to minimize the mass in this paper. For instance, our WE is based on the result reported by Kiewiet et al. [30] Their design shows decrease in mass-energy efficiency as input power to the reactor increases. In this paper, using a smaller number of reactors with higher input power (2.5 kW per reactor at maximum) is prioritized than mass-energy efficiency. Considering the same three metrics for more detailed plant design (e.g., number of reactors) should be included in the future.

Finally, Fig. 15(c) shows the Pareto optimal sets when only the mass payback ( $m_{O_2}/m_{sys}$ ) and mass-energy efficiency ( $m_{O_2}/E_{total}$ ) are considered. This highlights the clear benefits of deploying the WE architecture for both pilot and full-scale plants regardless of the scale and operational period of the full-scale plant since the other strategies are dominated by the WE strategy under various conditions. However, this also underscores the importance of considering more than two objectives to avoid decisions that might pose risks to other metrics. The impact of considering multiple objectives on the decision-making process should be explored further in future research.

##### 4.6.2. Carbothermal reduction reactant recycle rate

The system landed mass of the CR, PH, and SH architectures depends on the recycling rate of reactant for the CR process (i.e. methane and hydrogen) [20]. Fig. 16 shows two 3D Pareto fronts for different recycling rates. A comparison of Figs. 12 and 16 highlights the sensitivity of the CR and parallel strategies against this parameter. As the recycling rate becomes higher, the system mass that includes the CR process becomes smaller, which leads to a higher mass payback value. Since the reactant recycle rate does not affect the required energy or oxygen yield, the CR, parallel, and series strategies are only shifted along the mass payback axis direction.

When the recycling rate is 99%, the strategy of deploying a CR pilot and full-scale plant is expected to outperform the rest in terms of mass payback and oxygen yield (Fig. 17(a)). Furthermore, when considering the mass payback and the mass-energy efficiency (Fig. 17(b)), the WE strategy is no longer a single strategy that dominates the other strategies as seen in Fig. 13(c).

## 5. Discussion

### 5.1. Summary of key findings from the case study

The case study has discovered deploying either PH or SH pilot plant offers significant flexibility in decision-making. The Pareto optimal

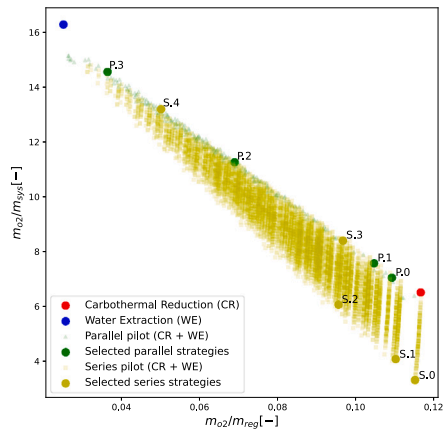
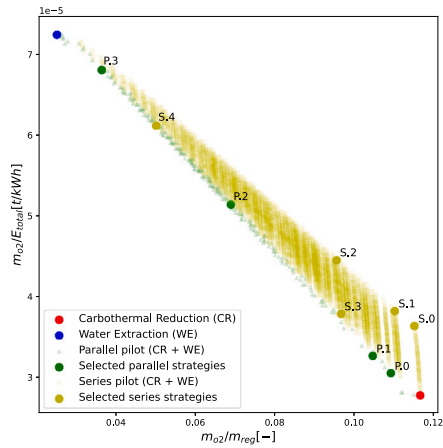
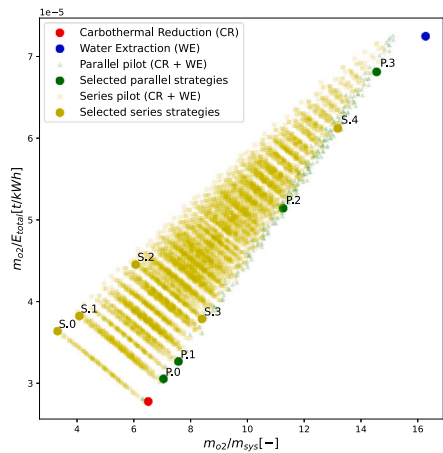
(a) Projection to oxygen yield ( $m_{O_2}/m_{reg}$ ) and mass payback ( $m_{O_2}/m_{sys}$ ) plane.(b) Projection to oxygen yield ( $m_{O_2}/m_{reg}$ ) and mass-energy efficiency ( $m_{O_2}/E_{total}$ ) plane.(c) Projection to mass payback ( $m_{O_2}/m_{sys}$ ) and mass-energy efficiency ( $m_{O_2}/E_{total}$ ) plane.

Fig. 13. Two dimensional projections of the Pareto front.

decision strategies (Fig. 12) involving PH or SH pilot plants are well-distributed in the three-dimensional objective space. This distribution suggests that using a hybrid pilot plant allows for adjustments to the strategy when deploying a full-scale plant, if necessary.

Furthermore, as Fig. 14 shows, deploying a CR or a WE plant in both pilot and full-scale is likely to work the best on one or two metrics

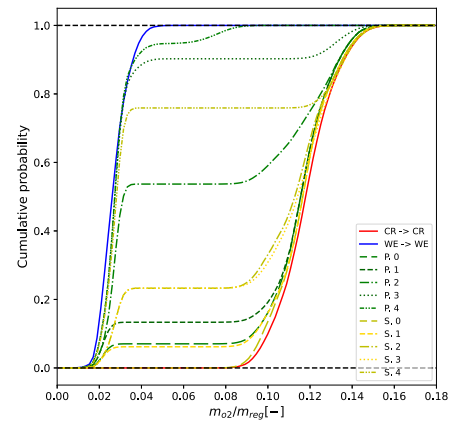
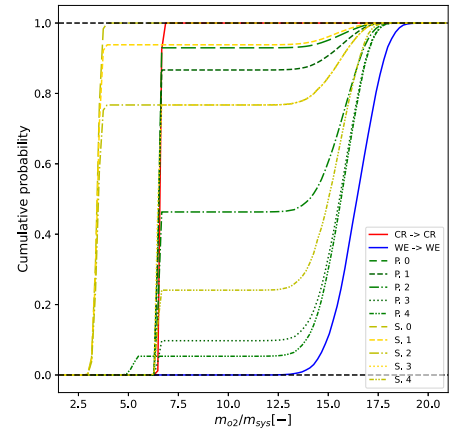
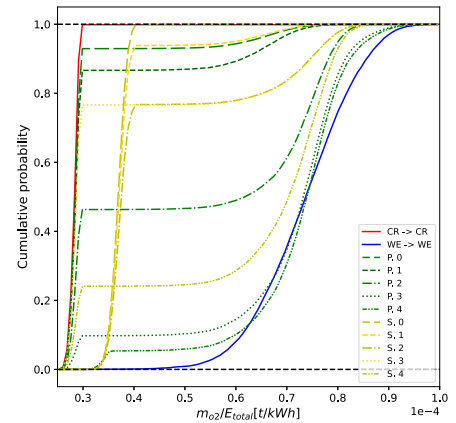
(a) Oxygen yield ( $m_{O_2}/m_{reg}$ ).(b) Mass payback ( $m_{O_2}/m_{sys}$ ).(c) Mass-energy efficiency ( $m_{O_2}/E_{total}$ ).

Fig. 14. Cumulative probabilities of each decision strategy.

while it could also work poorly for another metric. Many strategies with a hybrid pilot plant show more balanced performance for all three metrics. Cumulative probabilities of decision strategies that are not selected are examined in Appendix A.

Another key finding from the case study is that there is no single decision strategy that outperforms all others across all objectives. This highlights the importance of considering multiple objectives to avoid hidden risks in lunar ISRU operations.

In summary, if all three metrics are equally important, it is recommended to use a well-balanced hybrid strategy, such as P.2, S.3, or S.4.

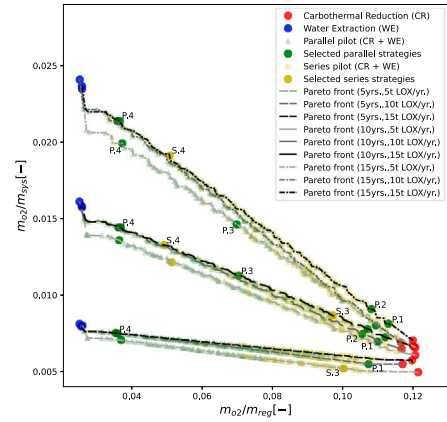
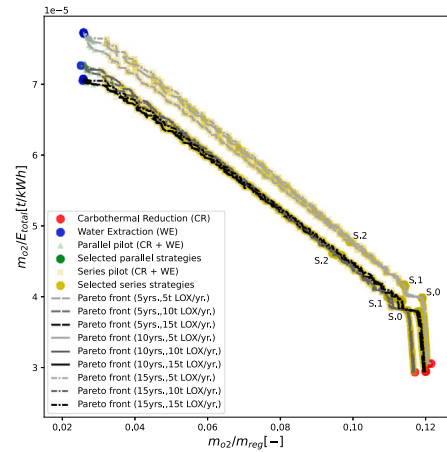
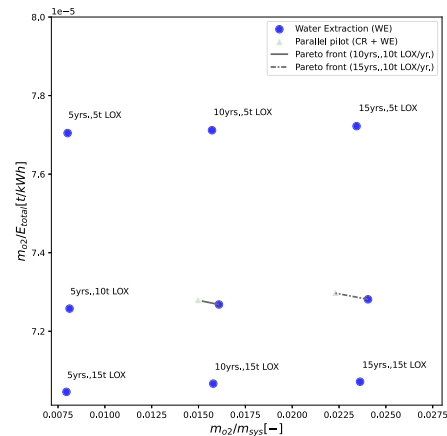
**Table 7**

Multi criteria table. Expected performance and its value with cumulative probability of 5% and 95%.

| (a) Oxygen yield ( $m_{O_2}/m_{reg}$ ) [%]. |                     |                     |                    |
|---------------------------------------------|---------------------|---------------------|--------------------|
| Strategy                                    | Expected            | 5% cumulative       | 95% cumulative     |
| CR                                          | 11.7                | 9.6                 | 14.1               |
| WE                                          | 2.5                 | 1.8                 | 3.9                |
| P.0                                         | 10.9                | 2.3                 | 13.9               |
| P.1                                         | 10.4                | 2.1                 | 13.9               |
| P.2                                         | 6.9                 | 2.0                 | 13.9               |
| P.3                                         | 3.6                 | 1.9                 | 13.2               |
| P.4                                         | 2.8                 | 1.9                 | 6.2                |
| S.0                                         | 11.5                | 9.3                 | 14.0               |
| S.1                                         | 11.0                | 2.4                 | 14.0               |
| S.2                                         | 8.7                 | 2.3                 | 14.0               |
| S.3                                         | 9.6                 | 2.3                 | 14.1               |
| S.4                                         | 5.1                 | 1.9                 | 14.1               |
| VoI                                         | −0.2<br>(CR vs S.0) | −0.3<br>(CR vs S.0) | 0.0<br>(CR vs S.3) |

| (b) Mass payback ( $m_{O_2}/m_{sys}$ ) [−]. |                     |                     |                     |
|---------------------------------------------|---------------------|---------------------|---------------------|
| Strategy                                    | Expected            | 5% cumulative       | 95% cumulative      |
| CR                                          | 6.5                 | 6.5                 | 6.7                 |
| WE                                          | 16.3                | 14.5                | 18.1                |
| P.0                                         | 7.0                 | 6.3                 | 14.9                |
| P.1                                         | 7.6                 | 6.3                 | 15.9                |
| P.2                                         | 11.3                | 6.3                 | 17.0                |
| P.3                                         | 14.6                | 6.5                 | 17.2                |
| P.4                                         | 15.0                | 5.5                 | 17.2                |
| S.0                                         | 3.3                 | 3.2                 | 3.7                 |
| S.1                                         | 4.1                 | 3.2                 | 14.4                |
| S.2                                         | 7.1                 | 3.2                 | 16.3                |
| S.3                                         | 8.4                 | 6.3                 | 16.3                |
| S.4                                         | 13.1                | 6.3                 | 16.9                |
| VoI                                         | −1.3<br>(WE vs P.4) | −8.0<br>(WE vs P.3) | −0.9<br>(WE vs P.4) |

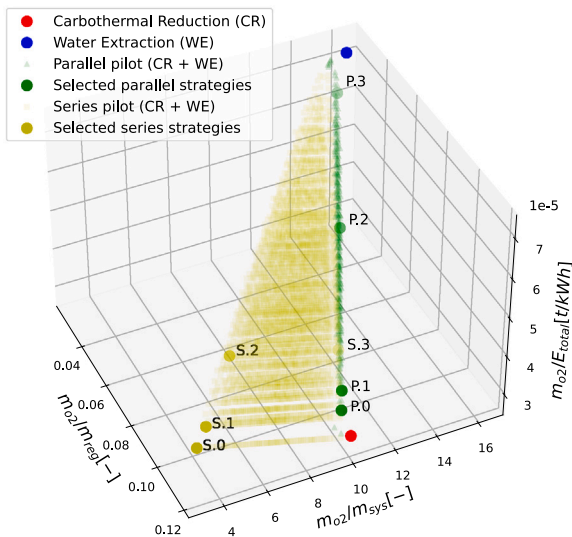
| (c) Mass-energy efficiency ( $m_{O_2}/E_{total}$ ) [ $\times 10^{-5}$ t/kWh] |                     |                     |                     |
|------------------------------------------------------------------------------|---------------------|---------------------|---------------------|
| Strategy                                                                     | Expected            | 5% cumulative       | 95% cumulative      |
| CR                                                                           | 2.8                 | 2.7                 | 3.0                 |
| WE                                                                           | 7.3                 | 5.8                 | 8.8                 |
| P.0                                                                          | 3.0                 | 2.7                 | 6.2                 |
| P.1                                                                          | 3.3                 | 2.7                 | 6.8                 |
| P.2                                                                          | 5.2                 | 2.7                 | 8.0                 |
| P.3                                                                          | 6.8                 | 2.8                 | 8.4                 |
| P.4                                                                          | 7.1                 | 3.5                 | 8.5                 |
| S.0                                                                          | 3.6                 | 3.4                 | 3.9                 |
| S.1                                                                          | 3.8                 | 3.4                 | 6.0                 |
| S.2                                                                          | 4.8                 | 3.5                 | 7.8                 |
| S.3                                                                          | 3.8                 | 2.7                 | 7.8                 |
| S.4                                                                          | 6.1                 | 2.8                 | 8.1                 |
| VoI                                                                          | −0.2<br>(WE vs P.4) | −2.3<br>(WE vs P.4) | −0.3<br>(WE vs P.3) |

(a) Oxygen yield ( $m_{O_2}/m_{reg}$ ) and mass payback ( $m_{O_2}/m_{sys}$ ) plane.(b) Oxygen yield ( $m_{O_2}/m_{reg}$ ) and mass-energy efficiency ( $m_{O_2}/E_{total}$ ) plane.(c) Mass payback ( $m_{O_2}/m_{sys}$ ) and mass-energy efficiency ( $m_{O_2}/E_{total}$ ) plane.**Fig. 15.** Two dimensional Pareto fronts under different assumptions.

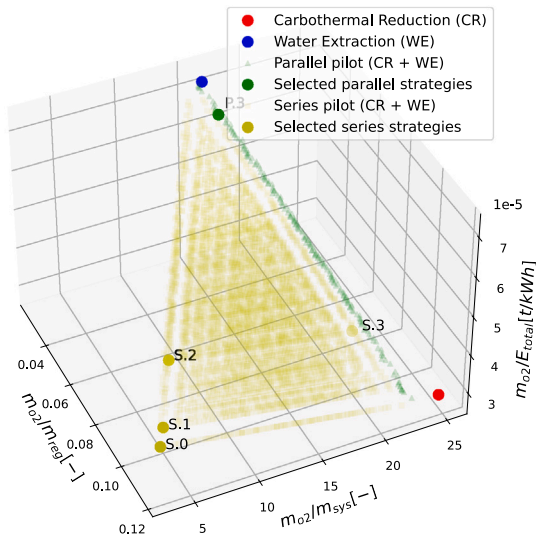
If oxygen yield is the primary criterion to consider, deploying a CR pilot and full-scale plant is a reasonable approach. On the other hand, when oxygen yield can be dismissed as a criterion, the WE strategy may be suitable. However, this leads to extensive excavation of valuable and sparse icy regolith, which may not be sustainable for future space exploration. For risk-averse decision-makers, it is recommended to deploy either a PH or SH pilot plant. Once more data is available and the criteria become clearer, a full-scale plant architecture can then be chosen with greater confidence following a strategy on the Pareto front.

## 5.2. Limitations of the study and direction for future work

One notable limitation of the proposed method is the high computational cost when the number of branches and stages increases. This limitation necessitated the simple case study with only two stages. For the baseline case study of this paper, the computation using Python 3.11 took approximately 11 min on a laptop with an Intel® Core™i7-1185G7 processor and 32 GB RAM.



(a) Recycling rate: 95%.



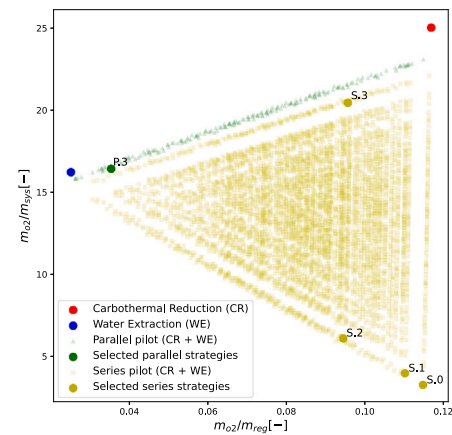
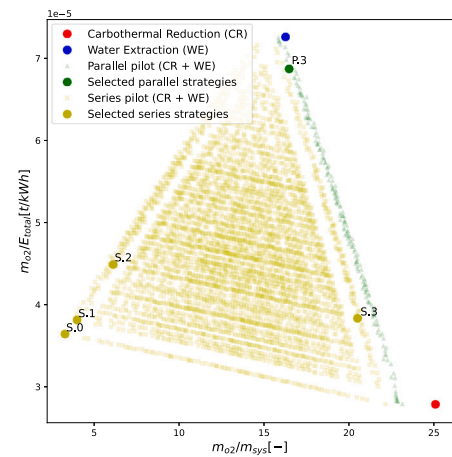
(b) Recycling rate: 99%.

**Fig. 16.** Three-dimensional Pareto fronts for different recycling rate. The production period and scale are set as 10 years and 10 t per annum, respectively.

For more complex scenarios, decision trees may no longer be suitable. While they work well for simple sequential decision problems, handling more complicated cases often requires either significant oversimplification or high computational costs.

The effect of ConOps and design parameters on the results should be explored in the future. In this paper, we simply employed the same architecture as explained by Ikeya et al. [20]. The optimal ConOps and design parameters for each architecture, however, might be different in terms of the three objectives considered. For instance, the type of excavation method (e.g., surface or subsurface), reactor geometry, and electrolysis (e.g., SOXE or proton exchange membrane electrolysis) affect the landed mass and energy consumption, and therefore, should be carefully chosen or included as other decisions in the decision tree.

Other oxygen extraction methods on the Moon, such as molten regolith electrolysis and molten salt electrolysis, could be included to

(a) Projection to oxygen yield ( $m_{O_2}/m_{reg}$ ) and mass payback ( $m_{O_2}/m_{sys}$ ) plane.(b) Projection to mass payback ( $m_{O_2}/m_{sys}$ ) and mass-energy efficiency ( $m_{O_2}/E_{total}$ ) plane.

**Fig. 17.** Two dimensional projections of the Pareto front for the case of 99% CR reactant recycling.

provide decision-makers with a more comprehensive assessment of the benefits and drawbacks of each architecture. Such inclusion can be done by using models developed by Guerrero-Gonzalez and Zabel [18].

Although oxygen is one of the most studied ISRU products, consideration of other products such as metals, silicon [18,70,81], and helium-3 [82,83] could give decision-makers a more comprehensive insight into lunar ISRU deployment. Incorporating Bayesian inference into the multi-commodity network model proposed and researched in the space logistics community (see e.g., [40]) would make a valuable invention for future cislunar logistics.

The conventional VoI approach is limited in that it cannot compare each decision across all considered objectives simultaneously. Therefore, there is a need to expand this approach to evaluate decisions within a multi-objective space.

Other flexible approaches should be explored further, too. In this paper, hybrid pilot plants are considered as an enabler of a flexible operation as they provide plenty of options for full-scale deployment. However, additional strategies could enhance the flexibility of ISRU architecture design and operation. For example, deploying multiple smaller-scale plants incrementally, instead of a single large-scale plant, could provide greater operational flexibility.

Finally, we recommend that our proposed method be combined with an economic model (see e.g., [84]) in the future. This combination and monetary assessments on VoI through case studies will provide space

agencies and private companies with better guidance on optimizing their project portfolios across various technology investments.

## 6. Conclusion

To address the challenge in decision-making in lunar ISRU plants under uncertainty, this paper has proposed using multi-objective Bayesian decision tree analysis. Multi-objective decision tree analysis is selected for its simplicity and its capability to gain insights into more complex problems. The proposed method combines traditional multi-objective decision tree analysis with Monte Carlo simulations and Bayesian inference. The method is used to identify optimal decision strategies of deploying lunar ISRU pilot and full-scale plants. The result indicates that, while no single strategy outperforms the rest in terms of all three metrics, mass payback, oxygen yield, and mass-energy efficiency, deploying the PH or SH in pilot scale seems like a well-balanced option. Deploying the WE architecture may be the best option for mass payback and oxygen yield; however, its relatively poor performance in oxygen yield indicates its risk for sustainable resource utilization. The produced Pareto front highlights the flexibility in deploying a hybrid architecture as a pilot plant, underscoring their adaptability to unknown requirements (e.g., in available energy) in the future.

Future work should explore the impact of the discrete state of nature in the decision tree, and assess alternative methods, such as stochastic programming, model predictive control, and reinforcement learning, for addressing more complex lunar ISRU deployment scenarios. A case study considering other ISRU technologies, such as molten salt electrolysis, as well as other ISRU products should be assessed to help decision-makers. Implementation of ConOps and design parameter optimizations into the decision-making process can be invaluable, too. Other methods to enhance flexible ISRU operations should also be explored. Finally, integrating our methodology with an ISRU economic model would provide space agencies and private companies with insights into how to optimally allocate their technology investment budgets.

## CRedit authorship contribution statement

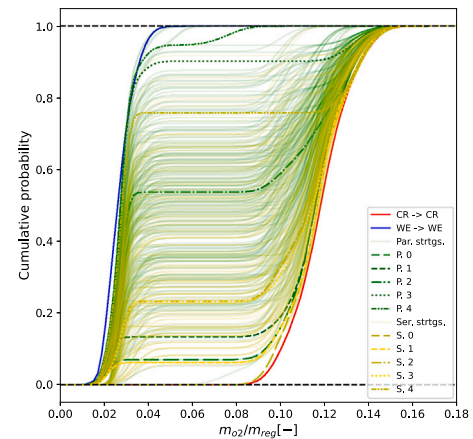
**Kosuke Ikeya:** Writing – original draft, Methodology, Formal analysis, Conceptualization, Visualization, Software, Investigation, Data curation, Writing – review & editing. **Michel-Alexandre Cardin:** Writing – review & editing, Validation, Supervision, Project administration, Resources, Methodology, Conceptualization. **Jan Cilliers:** Resources, Funding acquisition, Supervision, Conceptualization. **Stanley O. Starr:** Writing – review & editing, Supervision, Conceptualization. **Kathryn Hadler:** Supervision, Conceptualization.

## Declaration of competing interest

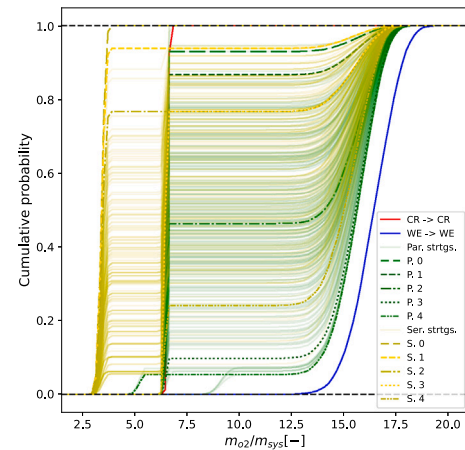
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgments

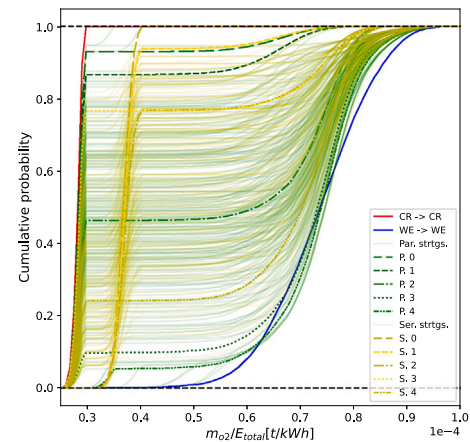
The authors are grateful for the financial support provided by the doctoral studentship scheme in the Department of Earth Science and Engineering at Imperial College London. We would like to thank Joshua Rasera and Luka Malone for their advice on modeling and other analytical aspects of the study.



(a) Oxygen yield ( $m_{O_2}/m_{reg}$ ).



(b) Mass payback ( $m_{O_2}/m_{sys}$ ).



(c) Mass-energy efficiency ( $m_{O_2}/E_{total}$ ).

Fig. A.1. Cumulative probabilities of decision strategies. Aside from the CR or the WE strategy, 200 strategies (strlgs.) are randomly undersampled from the Pareto front.

## Appendix A. Performance of decision strategies on the Pareto front

The Pareto front in Fig. 12 includes 245 and 6,736 strategies after the PH pilot and the SH pilot plant deployment. Fig. A.1 shows the cumulative probabilities of randomly undersampled parallel and series strategies (200 for each) on the Pareto front. Fig. A.1(a) shows that

the performance of almost any parallel or series strategies is between the CR and the WE strategies. Fig. A.1(b) shows the cost of the SH architecture in terms of its mass while Fig. A.1(c) shows its advantages in mass-energy efficiency.

## References

- [1] S. Creech, J. Guidi, D. Elburn, Artemis: an overview of NASA's activities to return humans to the moon, in: 2022 IEEE Aerospace Conference (AERO), Big Sky, Montana, 2022, pp. 1–7, <http://dx.doi.org/10.1109/AERO53065.2022.9843277>.
- [2] X. Wu, The international lunar research station: China's new era of space cooperation and its new role in the space legal order, *Space Policy* 65 (2023) 101537, <http://dx.doi.org/10.1016/j.spacepol.2022.101537>.
- [3] L. Schlüter, A. Cowley, Review of techniques for in-situ oxygen extraction on the moon, *Planet. Space Sci.* 181 (2020) 104753, <http://dx.doi.org/10.1016/j.pss.2019.104753>.
- [4] L.A. Taylor, W.D. Carrier, Production of oxygen on the moon: which processes are best and why, *AIAA J.* 30 (12) (1992) 2858–2863, <http://dx.doi.org/10.2514/3.48974>.
- [5] B.C. White, N.P. Haggerty, Carbothermal reduction system overview and developments in support of the artemis program and a commercial lunar economy, in: 52nd International Conference on Environmental Systems, Calgary, Alberta, Canada, 2023.
- [6] J. Prinetto, A. Colagrossi, A. Dottori, I. Troisi, M.R. Lavagna, Terrestrial demonstrator for a low-temperature carbothermal reduction process on lunar regolith simulant: design and AIV activities, *Planet. Space Sci.* 225 (2023) 105618, <http://dx.doi.org/10.1016/j.pss.2022.105618>.
- [7] K.A. Lee, L. Oryshchyn, A. Paz, M. Reddington, T.M. Simon, The roxygen project: outpost-scale lunar oxygen production system development at Johnson Space Center, *J. Aerosp. Eng.* 26 (1) (2013) 67–73, [http://dx.doi.org/10.1061/\(ASCE\)AS.1943-5525.0000230](http://dx.doi.org/10.1061/(ASCE)AS.1943-5525.0000230).
- [8] U. Hegde, R. Balasubramaniam, S. Gokoglu, Development and validation of a model for hydrogen reduction of JSC-1A, in: 47th AIAA Aerospace Sciences Meeting including The New Horizons Forum and Aerospace Exposition, Orlando, Florida, 2009, <http://dx.doi.org/10.2514/6.2009-1389>.
- [9] H.M. Sargeant, S.J. Barber, M. Anand, F.A.J. Abernethy, S. Sheridan, I.P. Wright, A.D. Morse, Hydrogen reduction of lunar samples in a static system for a water production demonstration on the moon, *Planet. Space Sci.* 205 (2021) 105287, <http://dx.doi.org/10.1016/j.pss.2021.105287>.
- [10] R. Balasubramaniam, U. Hegde, S. Gokoglu, Carbothermal processing of lunar regolith using methane, *AIP Conf. Proc.* 969 (1) (2008) 157–161, <http://dx.doi.org/10.1063/1.2844962>.
- [11] R. Balasubramaniam, S. Gokoglu, U. Hegde, The reduction of lunar regolith by carbothermal processing using methane, *Int. J. Miner. Process.* 96 (1) (2010) 54–61, <http://dx.doi.org/10.1016/j.minpro.2010.06.001>.
- [12] I. Troisi, P. Lunghi, M. Lavagna, Oxygen extraction from lunar dry regolith: thermodynamic numerical characterization of the carbothermal reduction, *Acta Astronaut.* 199 (2022) 113–124, <http://dx.doi.org/10.1016/j.actaastro.2022.07.021>.
- [13] J.E. Kleinhenz, A. Paz, Case studies for lunar ISRU systems utilizing polar water, in: ASCEND 2020, American Institute of Aeronautics and Astronautics, Virtual, 2020, <http://dx.doi.org/10.2514/6.2020-4042>.
- [14] D.L. Linne, J.E. Kleinhenz, A. Paz, Lunar water pilot plant conceptual design, in: ASCEND 2020, American Institute of Aeronautics and Astronautics, Virtual, 2020, <http://dx.doi.org/10.2514/6.2020-4236>.
- [15] G.F. Sowers, C.B. Dreyer, Ice mining in lunar Permanently Shadowed Regions, *New Space* 7 (4) (2019) 235–244, <http://dx.doi.org/10.1089/space.2019.0002>.
- [16] M. Anand, I.A. Crawford, M. Balat-Pichelin, S. Abanades, W. van Westrenen, G. Péraudeau, R. Jaumann, W. Seboldt, A brief review of chemical and mineralogical resources on the moon and likely initial in situ resource utilization (ISRU) applications, *Scientific Preparations For Lunar Exploration*, *Planet. Space Sci.* 74 (1) (2012) 42–48, <http://dx.doi.org/10.1016/j.pss.2012.08.012>.
- [17] H. Chen, T. Sartou du Jonchay, L. Hou, K. Ho, Integrated in-situ resource utilization system design and logistics for mars exploration, *Acta Astronaut.* 170 (2020) 80–92, <http://dx.doi.org/10.1016/j.actaastro.2020.01.031>.
- [18] F.J. Guerrero-Gonzalez, P. Zabel, System analysis of an ISRU production plant: extraction of metals and oxygen from lunar regolith, *Acta Astronaut.* 203 (2023) 187–201, <http://dx.doi.org/10.1016/j.actaastro.2022.11.050>.
- [19] A. Chepko, O. de Weck, W. Crossley, E. Santiago-Maldonado, D. Linne, A modeling framework for applying discrete optimization to system architecture selection and application to in-situ resource utilization, in: 12th AIAA/ISSMO Multidisciplinary Analysis and Optimization Conference, American Institute of Aeronautics and Astronautics, Victoria, British Columbia, Canada, 2008, <http://dx.doi.org/10.2514/6.2008-6058>.
- [20] K. Ikeya, F.J. Guerrero-Gonzalez, L. Kiewiet, M.-A. Cardin, J. Cilliers, S. Starr, K. Hadler, Hybrid lunar ISRU plant: A comparative analysis with carbothermal reduction and water extraction, *Acta Astronaut.* 230 (2025) 148–168, <http://dx.doi.org/10.1016/j.actaastro.2025.02.004>.
- [21] J.J. Cilliers, J.N. Rasera, K. Hadler, Estimating the scale of space resource utilisation (SRU) operations to satisfy lunar oxygen demand, *Planet. Space Sci.* 180 (2020) 104749, <http://dx.doi.org/10.1016/j.pss.2019.104749>.
- [22] Y.Y. Haimes, D. Li, V. Tulsiani, Multiobjective decision-tree analysis, *Risk Anal.* 10 (1) (1990) 111–127, <http://dx.doi.org/10.1111/j.1539-6924.1990.tb01026.x>.
- [23] S. Schreiner, L. Sibille, J. Dominguez, A. Sirk, J. Hoffman, G. Sanders, Development of a molten regolith electrolysis reactor model for lunar in-situ resource utilization in: 8th Symposium on Space Resource Utilization, American Institute of Aeronautics and Astronautics, Kissimmee, Florida, 2015, <http://dx.doi.org/10.2514/6.2015-1180>.
- [24] S.S. Schreiner, Molten Regolith Electrolysis Reactor Modeling and Optimization of In-Situ Resource Utilization Systems (Ph.D. thesis), Massachusetts Institute of Technology, 2015.
- [25] C. Schwandt, J.A. Hamilton, D.J. Fray, I.A. Crawford, The production of oxygen and metal from lunar regolith, *Scientific Preparations For Lunar Exploration*, *Planet. Space Sci.* 74 (1) (2012) 49–56, <http://dx.doi.org/10.1016/j.pss.2012.06.011>.
- [26] L. He, C. Wang, G. Zhang, Y. Pang, W. Yao, A novel auger-based system for extraterrestrial in-situ water resource extraction, *Icarus* 367 (2021) 114552, <http://dx.doi.org/10.1016/j.icarus.2021.114552>.
- [27] Y. Liu, C. Wang, Y. Pang, Q. Wang, Z. Zhao, T. Lin, Z. Wang, T. Shen, S. Liu, J. Song, X. Lai, X. Quan, W. Yao, Water extraction from icy lunar regolith by drilling-based thermal method in a pilot-scale unit, *Acta Astronaut.* 202 (2023) 386–399, <http://dx.doi.org/10.1016/j.actaastro.2022.11.002>.
- [28] K. Zacny, K. Luczek, A. Paz, M. Hedlund, Planetary Volatiles Extractor (PVEx) for In Situ Resource Utilization (ISRU), *American Society of Civil Engineers*, 2017, pp. 378–390, <http://dx.doi.org/10.1061/9780784479971.037>.
- [29] C. Purrington, G. Sowers, C. Dreyer, Thermal mining of volatiles in lunar regolith simulant, *Planet. Space Sci.* 222 (2022) 105550, <http://dx.doi.org/10.1016/j.pss.2022.105550>.
- [30] L. Kiewiet, N. Hab, F. Marchese, R. Freer, P. Zabel, Trade-off and optimization for a thermal lunar water extractor extraction system, in: 73rd International Astronautical Congress, Paris, France, 2022.
- [31] J. Cilliers, K. Hadler, J. Rasera, Toward the utilisation of resources in space: Knowledge gaps, open questions, and priorities, *Npj Microgravity* 9 (1) (2023) 1–5, <http://dx.doi.org/10.1038/s41526-023-00274-3>.
- [32] K. Hadler, D.J.P. Martin, J. Carpenter, J.J. Cilliers, A. Morse, S. Starr, J.N. Rasera, K. Seweryn, P. Reiss, A. Meurisse, A universal framework for space resource utilisation (SRU), *Planet. Space Sci.* 182 (2020) 104811, <http://dx.doi.org/10.1016/j.pss.2019.104811>.
- [33] L.S. Gertsch, R.E. Gertsch, Surface mine design and planning for lunar regolith production, *AIP Conf. Proc.* 654 (1) (2003) 1108–1115, <http://dx.doi.org/10.1063/1.1541408>.
- [34] T. Lager, A new conceptual model for the development of process technology in process industry: a point of departure for the transformation of the “process development process” into a formal work process?, *International Journal of Innovation Management* 04 (03) (2000) 319–346, <http://dx.doi.org/10.1142/S1363919600000159>.
- [35] L.P.L. Lim, E. Garnsey, M. Gregory, Product and process innovation in biopharmaceuticals: A new perspective on development, *R & D Manage.* 36 (1) (2006) 27–36, <http://dx.doi.org/10.1111/j.1467-9310.2006.00413.x>.
- [36] M. Kurkkio, J. Frishammar, U. Lichtenthaler, Where process development begins: A multiple case study of front end activities in process firms, *Technovation* 31 (9) (2011) 490–504, <http://dx.doi.org/10.1016/j.technovation.2011.05.004>.
- [37] B. Chen, J. He, X.-H. Wen, W. Chen, A.C. Reynolds, Uncertainty quantification and value of information assessment using proxies and Markov chain Monte Carlo method for a pilot project, *J. Pet. Sci. Eng.* 157 (2017) 328–339, <http://dx.doi.org/10.1016/j.petrol.2017.07.039>.
- [38] G.B. Sanders, W.E. Larson, Progress made in lunar in situ resource utilization under NASA's exploration technology and development program, in: *Earth and Space 2012*, American Society of Civil Engineers, Pasadena, California, 2012, pp. 457–478, <http://dx.doi.org/10.1061/9780784412190.050>.
- [39] A. Dottori, I. Troisi, M. Lavagna, s. Pirrotta, F. Latini, Lunar pilot plant payload design toward in situ demonstration of oxygen extraction by carbothermal reduction, 2023, <http://dx.doi.org/10.13009/EUCASS2023-283>.
- [40] H. Chen, T. Sartou du Jonchay, L. Hou, K. Ho, Multifidelity space mission planning and infrastructure design framework for space resource logistics, *J. Spacecr. Rockets* 58 (2) (2021) 538–551, <http://dx.doi.org/10.2514/1.A34666>.
- [41] Y. Takubo, H. Chen, K. Ho, Hierarchical reinforcement learning framework for stochastic spaceflight campaign design, *J. Spacecr. Rockets* 59 (2) (2022) 421–433, <http://dx.doi.org/10.2514/1.A35122>.
- [42] A. Babajide, R. de Neufville, M.-A. Cardin, Integrated method for designing valuable flexibility in oil development projects, *SPE Proj. Facil. Constr.* 4 (02) (2009) 3–12, <http://dx.doi.org/10.2118/122710-PA>.
- [43] S. Zuluaga, M. Sánchez-Silva, The value of flexibility and sequential decision-making in maintenance strategies of infrastructure systems, *Struct. Saf.* 84 (2020) 101916, <http://dx.doi.org/10.1016/j.strusafe.2019.101916>.
- [44] M. Samis, J. Steen, Financial evaluation of mining innovation pilot projects and the value of information, *Resour. Policy* 69 (2020) 101848, <http://dx.doi.org/10.1016/j.resourpol.2020.101848>.

- [45] W.B. Powell, Canonical problems and applications, in: Reinforcement Learning and Stochastic Optimization, John Wiley & Sons, Ltd, ISBN: 978-1-119-81506-8, 2022, pp. 39–100, <http://dx.doi.org/10.1002/9781119815068.ch2>.
- [46] R. Bellman, The theory of dynamic programming, Bull. Am. Math. Soc. 60 (6) (1954) 503–515, <http://dx.doi.org/10.1090/S0002-9904-1954-09848-8>.
- [47] M. Giuliani, J.D. Quinn, J.D. Herman, A. Castelletti, P.M. Reed, Scalable multiobjective control for large-scale water resources systems under uncertainty, IEEE Trans. Control Syst. Technol. 26 (4) (2018) 1492–1499, <http://dx.doi.org/10.1109/TCST.2017.2705162>.
- [48] M.-A. Cardin, Q. Xie, T.S. Ng, S. Wang, J. Hu, An approach for analyzing and managing flexibility in engineering systems design based on decision rules and multistage stochastic programming, IIEE Trans. 49 (1) (2017) 1–12, <http://dx.doi.org/10.1080/0740817X.2016.1189627>.
- [49] W. Wang, D.E. Rivera, Model predictive control for tactical decision-making in semiconductor manufacturing supply chain management, IEEE Trans. Control Syst. Technol. 16 (5) (2008) 841–855, <http://dx.doi.org/10.1109/TCST.2007.916327>.
- [50] R.G. Cestari, A. Castelletti, S. Formentin, Non-linear multi-objective Bayesian MPC of water reservoir systems, Eur. J. Control 83 (2025) 101205, <http://dx.doi.org/10.1016/j.ejcon.2025.101205>.
- [51] J.N. Morgan, J.A. Sonquist, Problems in the analysis of survey data, and a proposal, J. Amer. Statist. Assoc. 58 (302) (1963) 415–434, <http://dx.doi.org/10.1080/01621459.1963.10500855>.
- [52] T. Pelech, L. Yao, S. Saydam, Planning lunar in-situ resource utilisation with a reinforcement learning agent, Acta Astronaut. 201 (2022) 401–419, <http://dx.doi.org/10.1016/j.actaastro.2022.09.040>.
- [53] C. Yüksel, C. Minnecker, M.S. Shishvan, J. Benndorf, M. Buxton, Value of information introduced by a resource model updating framework, Math. Geosci. 51 (7) (2019) 925–943, <http://dx.doi.org/10.1007/s11004-018-9770-x>.
- [54] M.-A. Cardin, S.J. Steer, W.J. Nuttall, G.T. Parks, L.V.N. Gonçalves, R. de Neufville, Minimizing the economic cost and risk to accelerator-driven subcritical reactor technology. part 2: the case of designing for flexibility, Nucl. Eng. Des. 243 (2012) 120–134, <http://dx.doi.org/10.1016/j.nucengdes.2011.11.026>.
- [55] R. de Neufville, S. Scholtes, Flexibility in engineering design, 2011, <http://dx.doi.org/10.7551/mitpress/8292.001.0001>.
- [56] M.-A. Cardin, A. Mijic, J. Whyte, Flexibility and real options in engineering systems design, ISBN: 978-3-030-46054-9, 2022, [http://dx.doi.org/10.1007/978-3-030-46054-9\\_35-1](http://dx.doi.org/10.1007/978-3-030-46054-9_35-1).
- [57] E.M. Coopersmith, P.C. Cunningham, A practical approach to evaluating the value of information and real option decisions in the upstream petroleum industry, in: SPE Annual Technical Conference and Exhibition, OnePetro, San Antonio, Texas, 2002, <http://dx.doi.org/10.2118/77582-MS>.
- [58] G.A. Davis, M. Samis, Using real options to value and manage exploration, in: M.D. Doggett, J.R. Parry (Eds.), in: Wealth Creation in the Minerals Industry: Integrating Science, Business, and Education, vol. 12, Society of Economic Geologists, ISBN: 978-1-62949-036-6, 2005, <http://dx.doi.org/10.1180/EMU-notes.11.1>.
- [59] L. Trigeorgis, Real Options Managerial Flexibility and Strategy in Resource Allocation, MIT Press, 1996, <https://mitpress.mit.edu/9780262201025/real-options/>.
- [60] N. Huntley, M.C.M. Trofaes, Normal form backward induction for decision trees with coherent lower previsions, Ann. Oper. Res. 195 (1) (2012) 111–134, <http://dx.doi.org/10.1007/s10479-011-0968-2>.
- [61] R. Bellman, Dynamic Programming, Princeton University Press, 1957.
- [62] A. Frini, A. Guitouni, J.-M. Martel, A general decomposition approach for multi-criteria decision trees, European J. Oper. Res. 220 (2) (2012) 452–460, <http://dx.doi.org/10.1016/j.ejor.2012.01.032>.
- [63] E. Mazarico, G.A. Neumann, D.E. Smith, M.T. Zuber, M.H. Torrence, Illumination conditions of the Lunar Polar Regions using LOLA topography, Icarus 211 (2) (2011) 1066–1081, <http://dx.doi.org/10.1016/j.icarus.2010.10.030>.
- [64] A. Colaprete, P. Schultz, J. Heldmann, D. Wooden, M. Shirley, K. Ennico, B. Hermalyn, W. Marshall, A. Ricco, R.C. Elphic, D. Goldstein, D. Summy, G.D. Bart, E. Asphaug, D. Korycansky, D. Landis, L. Sollitt, Detection of water in the LCROSS ejecta plume, Science 330 (6003) (2010) 463–468, <http://dx.doi.org/10.1126/science.1186986>.
- [65] C.A. Severi, V. Pérez, C. Pascual, R. Muñoz, R. Lebrero, Identification of critical operational hazards in a biogas upgrading pilot plant through a multi-criteria decision-making and FTOPSIS-HAZOP approach, Chemosphere 307 (2022) 135845, <http://dx.doi.org/10.1016/j.chemosphere.2022.135845>.
- [66] G. Sanders, J. Kleinhenz, D. Linne, NASA plans for in situ resource utilization (ISRU) development, demonstration, and implementation, in: Committee on Space Research (COSPAR) 2022, Athens, Greece, 2022.
- [67] R.P. Mueller, J.D. Smith, J.M. Schuler, A.J. Nick, N.J. Gelino, K.W. Leucht, I.I. Townsend, A.G. Dokos, Design of an excavation robot: regolith advanced surface systems operations robot (RASSOR) 2.0, in: Earth and Space 2016, American Society of Civil Engineers, Orlando, Florida, 2017, pp. 163–174, <http://dx.doi.org/10.1061/9780784479971.018>.
- [68] E. Rice, S. Rosenberg, O. Musbah, P. Hermes, P. Bemowski, Carbothermal reduction of lunar materials for oxygen production on the moon, in: 34th Aerospace Sciences Meeting and Exhibit, American Institute of Aeronautics and Astronautics, Reno, Nevada, 1996, <http://dx.doi.org/10.2514/6.1996-487>.
- [69] H.-C. Lee, S. Dhage, M.S. Akhtar, D.H. Kwak, W.J. Lee, C.-Y. Kim, O.-B. Yang, A simulation study on the direct carbothermal reduction of SiO<sub>2</sub> for Si metal, Curr. Appl. Phys. 10 (2) (2010) S218–S221, <http://dx.doi.org/10.1016/j.cap.2009.11.053>.
- [70] S. Kaur, W.J. Ready, T.M. Orlando, P.G. Loutzenhiser, Metal and metalloids production from lunar regolith simulants via carbothermal reduction: thermodynamic and experimental analyses, Acta Astronaut. 232 (2025) 479–492, <http://dx.doi.org/10.1016/j.actaastro.2025.03.021>.
- [71] S. Sathyan, M. Bhatt, M. Chowdhury, P. Gläser, D. Misra, N. Srivastava, S. Narendranath, K.S. Sajinkumar, A. Bhardwaj, Potential landing sites characterization on lunar south pole: de-gerlache to shackleton ridge region, Icarus 412 (2024) 115988, <http://dx.doi.org/10.1016/j.icarus.2024.115988>.
- [72] D. Kaschubek, M. Killian, L. Grill, System analysis of a moon base at the south pole: considering landing sites, ECLSS and ISRU, Acta Astronaut. 186 (2021) 33–49, <http://dx.doi.org/10.1016/j.actaastro.2021.05.004>.
- [73] A. Chepko, O. de Weck, D. Linne, E. Santiago-Maldonado, W. Crossley, Architecture modeling of in-situ oxygen production and its impacts on lunar campaigns, in: AIAA SPACE 2008 Conference & Exposition, in: AIAA SPACE Forum, American Institute of Aeronautics and Astronautics, San Diego, California, 2008, <http://dx.doi.org/10.2514/6.2008-7823>.
- [74] E. Hinterman, K. Carroll, A. Nikolic, O. de Weck, J. Hoffman, Multi-objective system optimization of a mars atmospheric ISRU plant for oxygen production, in: 2021 IEEE Aerospace Conference (50100), Virtual, 2021, pp. 1–12, <http://dx.doi.org/10.1109/AERO50100.2021.9438331>.
- [75] J.C. Graf, Lunar Soils Grain Size Catalog, NASA Reference Publication 1265, National Aeronautics and Space Administration, 1993.
- [76] W.I. Ridley, C. Donaldson, A.M. Reid, J.L. Warner, R. Gooley, R.W. Brown, Glass compositions in apollo 16 soils 60501 and 61221, in: Proceedings of the Fourth Lunar Science Conference, 1, 1973, pp. 309–321.
- [77] P.O. Hayne, A. Hendrix, E. Sefton-Nash, M.A. Siegler, P.G. Lucey, K.D. Retherford, J.-P. Williams, B.T. Greenhagen, D.A. Paige, Evidence for exposed water ice in the moon's South Polar Regions from lunar reconnaissance orbiter ultraviolet albedo and temperature measurements, Lunar Volatiles, Icarus 255 (2015) 58–69, <http://dx.doi.org/10.1016/j.icarus.2015.03.032>.
- [78] J.D. Cole, S. Lim, H.M. Sargeant, S. Sheridan, M. Anand, A. Morse, Water extraction from icy lunar simulants using low power microwave heating, Acta Astronaut. 209 (2023) 95–103, <http://dx.doi.org/10.1016/j.actaastro.2023.04.035>.
- [79] F.J. Guerrero-Gonzalez, C.F. Dietrich, F. Schuster, Isrilib documentation site, 2023, <https://collab.dvb.bayern/display/TUMIpe/ISRUlib>.
- [80] M. Lavagna, I. Troisi, A. Dottori, A. Colagrossi, J. Prinetto, R. Freddi, I. Pretto, Solid-gas carbothermal reduction for water production from lunar regolith simulant: Experimental results, in: 11th European Lunar Symposium, Padua, Italy, 2023, [https://www.youtube.com/watch?v=jm-0\\_k2nsJc](https://www.youtube.com/watch?v=jm-0_k2nsJc).
- [81] B. Thibodeau, X. Walls, A. Ellery, Extraction of silica and alumina from lunar highland simulant, in: Earth and Space 2024, American Society of Civil Engineers, Miami, Florida, 2024, pp. 260–268, <http://dx.doi.org/10.1061/9780784485736.024>.
- [82] C. Ding, Q. Li, J. Xu, Z. Lei, J. Li, Y. Su, S. Huang, Moon-based ground penetrating radar derivation of the helium-3 reservoir in the regolith at the chang'e-3 landing site, IEEE J. Sel. Top. Appl. Earth Obs. Remote. Sens. 16 (2023) 2764–2776, <http://dx.doi.org/10.1109/JSTARS.2023.3253499>.
- [83] A.D. Olson, Lunar helium-3: mining concepts, extraction research, and potential ISRU synergies, in: ASCEND 2021, American Institute of Aeronautics and Astronautics, Las Vegas, Nevada & Virtual, ISBN: 978-1-62410-612-5, 2021, <http://dx.doi.org/10.2514/6.2021-4237>.
- [84] P.T. Metzger, Economics of in-space industry and competitiveness of lunar-derived rocket propellant, Acta Astronaut. 207 (2023) 425–444, <http://dx.doi.org/10.1016/j.actaastro.2023.03.014>.